



[INSTITUTION NAME]

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CAPSTONE PROJECT REPORT

Farmer Groceries

*A Direct Farmer-to-Consumer Grocery Marketplace
with Role-Based Mobile Workflows for Customers, Farmers and Delivery Partners*

Submitted in partial fulfilment of the requirements for the award of the degree of
Bachelor of Science in Computer Science (Online Mode)

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Highlighted fields are to be completed before submission.

DECLARATION

I/We hereby declare that this capstone project titled “**Farmer Groceries — A Direct Farmer-to-Consumer Grocery Marketplace with Role-Based Mobile Workflows**” is an original work carried out by me/us under the guidance of *[Supervisor Name]*, and that it has not been submitted to any other university or institution for the award of any degree, diploma or other similar title.

All sources of information, third-party libraries, frameworks and reference material used in the course of this work have been duly acknowledged in the References section of this report. The source code accompanying this report is maintained under version control and is available in the public repository cited on the cover page. Any content reproduced from other works is cited explicitly, and the implementation, testing and documentation presented here represent our own effort.

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ABSTRACT

The Indian agricultural supply chain routinely places five or more intermediaries between the grower and the household that eventually consumes the produce. Each intermediary adds margin without adding value, with the result that farmers realise only a fraction of the final retail price while consumers pay a substantial mark-up, and perishable produce loses freshness at every additional hop. **Farmer Groceries** is a cross-platform mobile application that compresses this chain to a single link — farmer to delivery partner to consumer — inside one application serving three distinct user roles.

The implemented system is a React Native application built on the Expo SDK 54 managed workflow, written entirely in TypeScript under strict compiler settings and routed with the file-based Expo Router. A single authenticated identity is specialised at sign-in time into one of three role workspaces: customers browse a categorised catalogue, manage a persistent cart and track orders through a seven-stage progress timeline; farmers publish and price inventory, monitor low-stock alerts and advance incoming orders; delivery partners receive nearby jobs with distance-derived ETAs, accept or decline them, follow a pickup-to-drop route view and monitor period-wise earnings. Global state is held in React Context providers for authentication and cart, while all persistence is funnelled through a single service façade that dispatches either to a Supabase (PostgreSQL + Auth + Storage) backend or, when no cloud credentials are configured, to an on-device data layer persisted through AsyncStorage. This dual-target design makes the application offline-capable — every screen continues to function without connectivity — while remaining ready for multi-tenant cloud deployment.

The delivered build comprises 48 TypeScript source files (approximately 5,400 lines) spanning 18 routed screens and a 12-component reusable design system driven by a central token module. The application compiles cleanly under `tsc --noEmit` in strict mode and was validated by automated end-to-end suites, committed to the repository, that drive all three role workflows through a real browser — covering sign-in, role selection, catalogue browsing, cart and checkout, order placement, farmer-side fulfilment and delivery acceptance through to completion. Forty functional and validation test cases were executed, of which all forty passed. The outcome is a working, demonstrable marketplace that measurably shortens the produce supply chain and provides a credible foundation for a production deployment.

Keywords

Agricultural supply chain; disintermediation; React Native; Expo; TypeScript; Supabase; role-based access; mobile commerce; order lifecycle management; last-mile delivery.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AAB	Android App Bundle
API	Application Programming Interface
APK	Android Package Kit
CDP	Chrome DevTools Protocol
COD	Cash on Delivery
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DOM	Document Object Model
EAS	Expo Application Services
ER	Entity Relationship
ETA	Estimated Time of Arrival
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVP	Minimum Viable Product
npm	Node Package Manager
PDF	Portable Document Format
RLS	Row Level Security
RN	React Native
SDG	Sustainable Development Goal
SDK	Software Development Kit
SQL	Structured Query Language
UI / UX	User Interface / User Experience
UPI	Unified Payments Interface
URL	Uniform Resource Locator
UUID	Universally Unique Identifier

CHAPTER 1: INTRODUCTION

Note: Problem identification and system design were completed and approved as part of the preceding Study Project. This chapter reproduces that work in condensed form and records the refinements made during implementation. The principal changes are: (i) the adoption of Supabase in place of the originally proposed Firebase backend, (ii) the addition of an on-device data layer so that the application remains fully functional without network connectivity, and (iii) the extension of the order lifecycle from five to seven states to give delivery partners explicit *assigned* and *picked up* checkpoints.

1.1 Overview of the Project

Farmer Groceries is a cross-platform mobile marketplace that connects agricultural producers directly with the households that buy their produce, and coordinates the last-mile delivery between them. The application is built as a single React Native codebase that runs on Android, iOS and the web, and it serves three distinct classes of user from that one codebase. A person signs up once, chooses a role, and is thereafter routed into a workspace tailored to that role: a shopping experience for customers, an inventory and fulfilment console for farmers, and a job board with route and earnings tracking for delivery partners.

The commercial premise of the system is disintermediation. Every order placed in the application creates a direct commercial relationship between one farmer and one customer, with a delivery partner compensated separately for the physical movement of goods. There is no wholesaler, commission agent or retailer in the transaction path, so the margin those parties would otherwise extract is redistributed between the producer and the consumer.

The delivered artefact is a complete, runnable application rather than a prototype of selected screens. All eighteen routed screens are implemented and reachable, the order lifecycle can be driven from placement to delivery by three different users acting in sequence, and state changes made by one role are immediately visible to the others. The system persists data through a service layer that can be pointed either at a hosted Supabase project or at the on-device data layer, so the application keeps working whether or not the network does.

1.2 Problem Statement and Motivation

Produce in the conventional Indian agricultural supply chain typically passes through five or more intermediaries between the field and the kitchen. A grower sells to a local trader or an aggregator, who sells to a commission agent operating in a regulated wholesale market, who sells to a wholesaler, who sells to a retailer, who finally sells to the household. Each hop adds

a margin, a handling delay and an opportunity for spoilage, and none of the hops after the first adds any agronomic value to the product.

Three consequences follow directly from this structure, and together they constitute the problem this project addresses:

- **Producer income is compressed.** The share of the final retail price that reaches the grower is a small fraction of what the consumer pays, because each intermediate party prices for its own risk and working capital rather than for value added.
- **Consumer prices are inflated.** The household pays for a chain of margins it never interacts with, and has no visibility into who grew the produce, when it was harvested, or under what conditions.
- **Freshness and quantity are lost.** Perishables spend days in transit and storage across the chain. Wastage accumulates at every transfer, and what survives arrives materially less fresh than it left the farm.

The structural fix is to shorten the chain, and the practical obstacle to shortening it is coordination. A farmer cannot economically discover, price for, and deliver to individual households without software; a household cannot discover nearby farmers or verify their produce without a catalogue; and neither party can arrange reliable last-mile logistics alone. A single application that carries all three participants and mediates the transaction between them removes that coordination barrier. Figure 1.1 contrasts the two structures.

(a) Conventional chain — five intermediaries



Margin accrues at every hop · handling delay compounds · perishables degrade in transit

(b) Proposed chain — one coordinating application

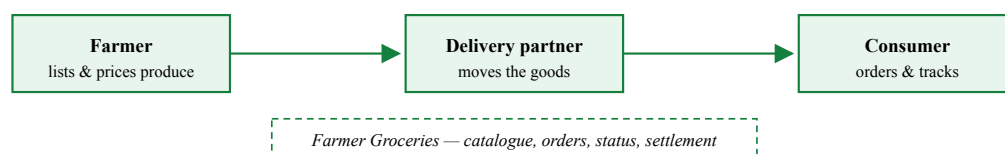


Figure 1.1 — Conventional multi-intermediary produce chain versus the proposed direct chain mediated by a single application.

1.3 Objectives of the Capstone

Where the Study Project established *what* should be built and *why*, the capstone phase was concerned with building it, proving that it works, and documenting the result. Six objectives were set at the start of the implementation phase; Table 1.1 states each objective alongside the

concrete deliverable by which it is judged and the section of this report in which the evidence appears.

#	Objective	Deliverable used as evidence	Evidence in
O1	Implement a single authenticated identity that specialises into three role-specific workspaces with enforced navigation boundaries.	Auth context, persisted role, and a root navigation guard that redirects any user found outside their role's route group.	§2.3.1, §2.4.1
O2	Deliver a complete customer purchase journey from catalogue discovery to order tracking.	Seven customer screens including search, category and organic filters, persistent cart, checkout with address validation, and a seven-stage tracking timeline.	§2.3.2, §2.5
O3	Deliver a farmer console for inventory publication and order fulfilment.	Dashboard with revenue and low-stock analytics, product creation and visibility toggling, and an order queue with guarded status transitions.	§2.3.3, §2.5
O4	Deliver a delivery-partner workflow covering job acceptance, execution and remuneration.	Availability toggle, distance-ranked job board with computed ETAs, accept/decline actions, pickup-to-drop route view and period-wise earnings.	§2.3.4, §2.5
O5	Centralise persistence behind one interface so the backend can be substituted without touching screen code.	A service façade of thirty operations, each dispatching to Supabase or to the on-device data layer based on a single configuration flag.	§2.1.3, §2.4.5
O6	Validate the integrated system end to end and record reproducible evidence of correctness.	Strict-mode type check with zero errors, plus two committed browser-driven suites exercising all three roles across forty test cases.	Chapter 3

Table 1.1 — Objectives of the capstone and their realisation.

1.4 Scope of Implementation

The capstone scope was fixed deliberately at a functionally complete marketplace with simulated payment and simulated geolocation. This boundary was chosen because payment gateway integration and live location streaming both require commercial accounts and device permissions that add integration risk without demonstrating additional software-engineering competence. Every feature inside the boundary is implemented and exercised; every feature outside it is recorded in §6.4 as future work rather than silently omitted.

In scope — implemented and verified	Out of scope — deferred to future work
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• Email and password authentication with role assignment and session persistence • Role-guarded navigation across three route groups • Product catalogue over eight categories with search, category, organic and price sorting • Cart with stock-capped quantities, persistence across restarts and free-delivery progress • Checkout with address validation, three payment methods and per-farmer order splitting • Seven-stage order lifecycle with an immutable status history • Customer-initiated cancellation restricted to pre-fulfilment states • Farmer inventory CRUD, low-stock alerting and revenue aggregation • Delivery job board with Haversine distance and ETA, accept/decline, and route view • Period-wise earnings computation for delivery partners • Reusable design-token-driven component library and responsive web shell	• Live payment gateway settlement — payment status is recorded, not charged • GPS location streaming and turn-by-turn map navigation • Push notifications for order state changes • In-app messaging between customer and delivery partner • Product image upload — the storage path exists but the catalogue renders category glyphs • Customer-authored reviews and ratings — the data model and service methods exist, the authoring screen does not • Recurring subscription orders — modelled in the type system only • Administrative moderation and farmer document verification workflow • Multi-language localisation
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Table 1.2 — Scope boundaries: included and excluded functionality.

1.5 Organisation of the Report

Chapter 2 presents the implementation in detail: the layered architecture and its data flows, the technology stack and the reasoning behind each choice, a module-by-module description of the three role workspaces, the algorithms that encode the business rules, and annotated screenshots of the running application accompanied by the source listings that produce them.

Chapter 3 sets out the test strategy, the tooling used to execute it, the full set of functional and negative test cases with their recorded inputs and outcomes, and an analysis of the results including non-functional measurements.

Chapter 4 documents the execution environment, the steps required to run the application locally and to produce release builds for each target platform, and the demonstration material.

Chapter 5 provides project execution evidence: the version control history, a weekly progress summary derived from that history, and the review and feedback log.

Chapter 6 concludes with a summary of what was built, an honest assessment of achievements against the objectives of §1.3, the limitations of the current build, and a prioritised roadmap of enhancements. References and four appendices — a user manual, an installation guide, the source code location and the demonstration video link — complete the report.

CHAPTER 2: IMPLEMENTATION DETAILS

2.1 System Architecture and Design

2.1.1 High-Level Architecture

The application is organised into five layers with a strict downward dependency rule: a layer may import from the layers beneath it but never from those above. This constraint is what makes the backend substitutable — screens never import a database client, they import a service, and the service decides where the data comes from.

The **presentation layer** consists of eighteen routed screens declared as files under `app/`. Expo Router derives the navigation graph from the directory structure, so the three parenthesised directories `(customer)`, `(farmer)` and `(delivery)` become three independent tab stacks without any imperative navigator configuration. The **component layer** supplies twelve presentational primitives — buttons, inputs, chips, status pills, skeletons, toasts and a custom tab bar — all of which read their colours, radii and shadows from a single design-token module, so a change to the brand palette propagates through every screen at once.

The **state layer** comprises two React Context providers mounted at the root. `AuthProvider` owns the authenticated user and the selected role and exposes `setRole`, `signOut` and `refreshUser`; `CartProvider` owns the cart array and derives `cartTotal` and `cartCount` from it. Both hydrate from device storage on mount and write back on change, which is why a cart survives an application restart.

The **service layer** is a single class, `SupabaseService`, exposing thirty static operations grouped into authentication, users, farmers, products, orders, reviews and file upload. Every method begins with the same guard: if the environment supplies a Supabase URL the method issues a PostgREST query and maps the `snake_case` row into a `camelCase` domain object; if it does not, the method delegates to `localStore`. Neither branch leaks into the caller.

The **persistence layer** therefore has two interchangeable implementations. The remote implementation is a hosted Supabase project providing PostgreSQL storage, GoTrue authentication and object storage. The local implementation is an in-memory state object holding the catalogue, farmer profiles and order history, mirrored into `AsyncStorage` after every mutation and rehydrated — including date revival — on the next launch.

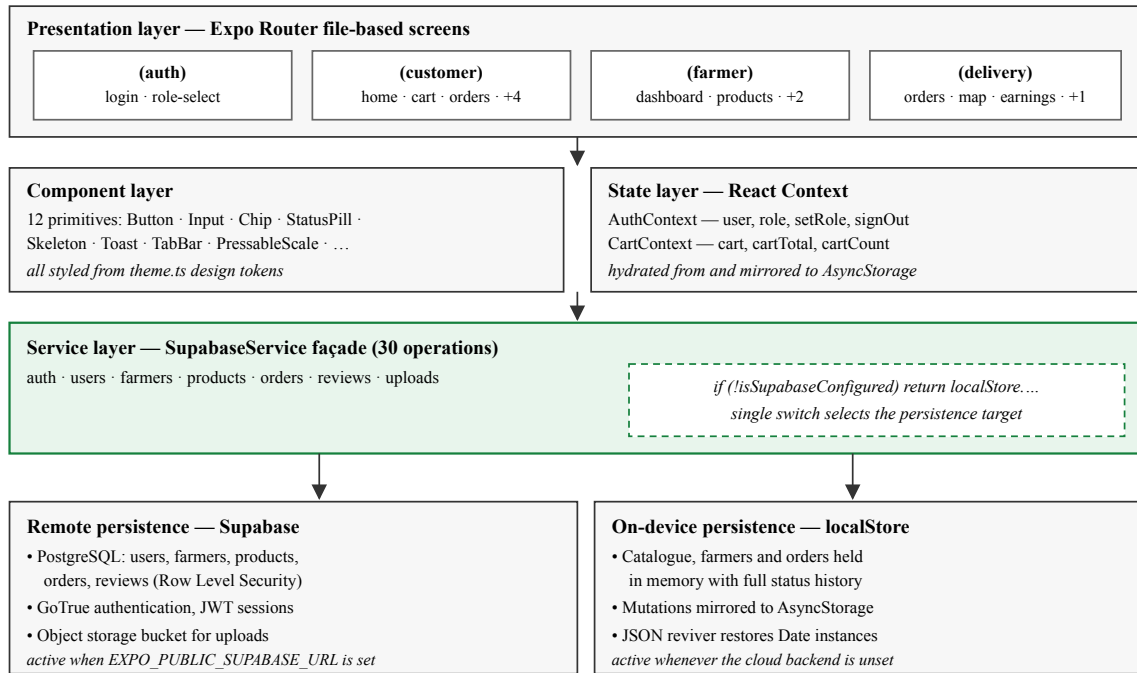


Figure 2.1 — High-level layered architecture. Dependencies point downward only; the service façade is the sole boundary between screens and storage.

2.1.2 Data Flow Diagrams

Figure 2.2 gives the context diagram. The system is treated as a single process with three external entities — the three human roles — and one external data service. Figure 2.3 decomposes that process into the five functional processes that the implementation actually contains, together with the four logical data stores they read and write.

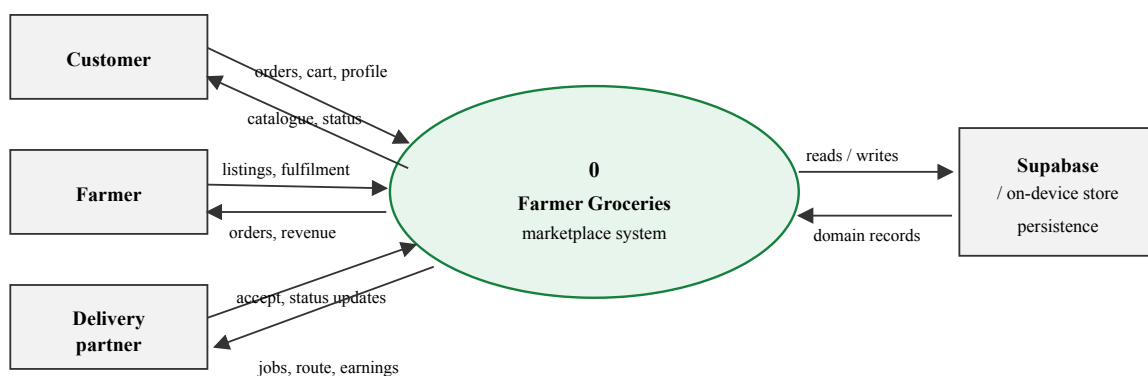


Figure 2.2 — Context diagram (Level-0 data flow diagram).

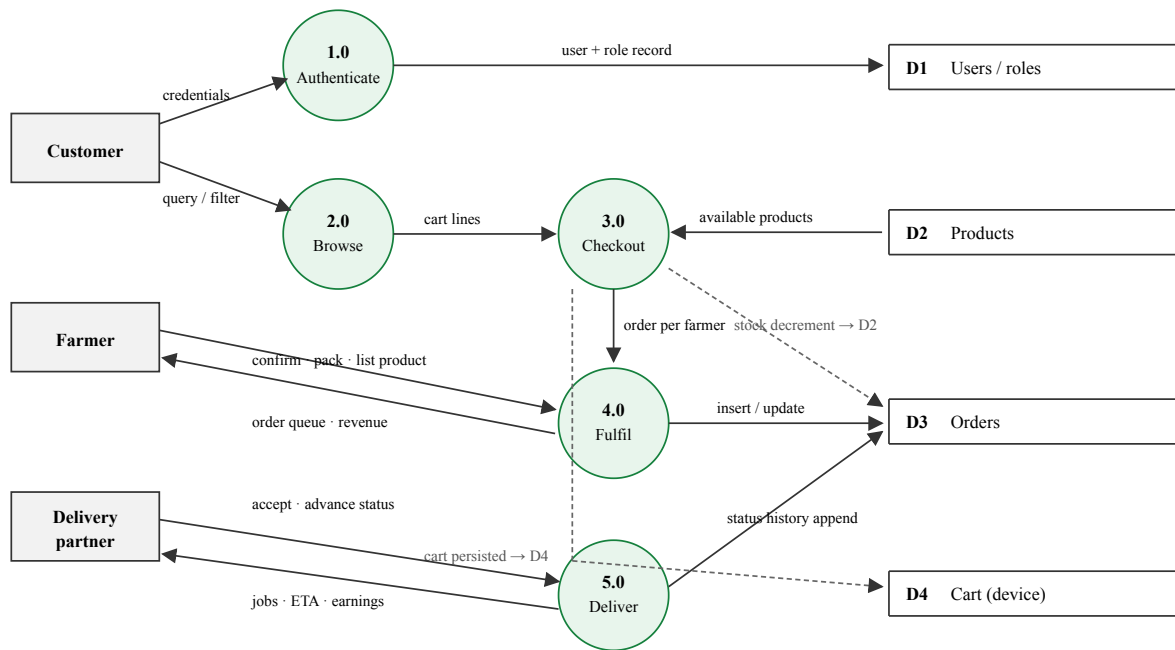


Figure 2.3 — Level-1 data flow diagram showing the five functional processes and the four logical data stores.

2.1.3 Component Interaction

Order placement is the interaction that touches the greatest number of components, and it is worth tracing in full. When the customer presses *Place Order*, the payment screen validates the cart total against the minimum order value and the address against the pincode pattern, groups the cart lines by `farmerId`, and issues one `createOrder` call per distinct farmer in parallel. The service façade routes each call to the active persistence target, which assigns an identifier, appends the initial `placed` status entry and decrements the stock of each ordered product. Only when every promise resolves is the cart cleared and the customer redirected to the order history. Figure 2.4 shows the sequence.

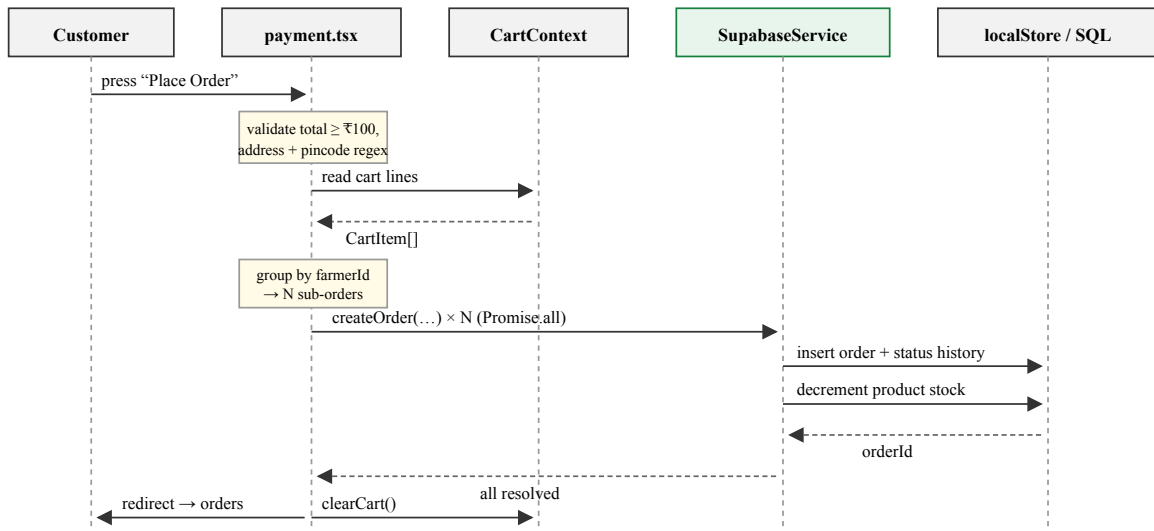
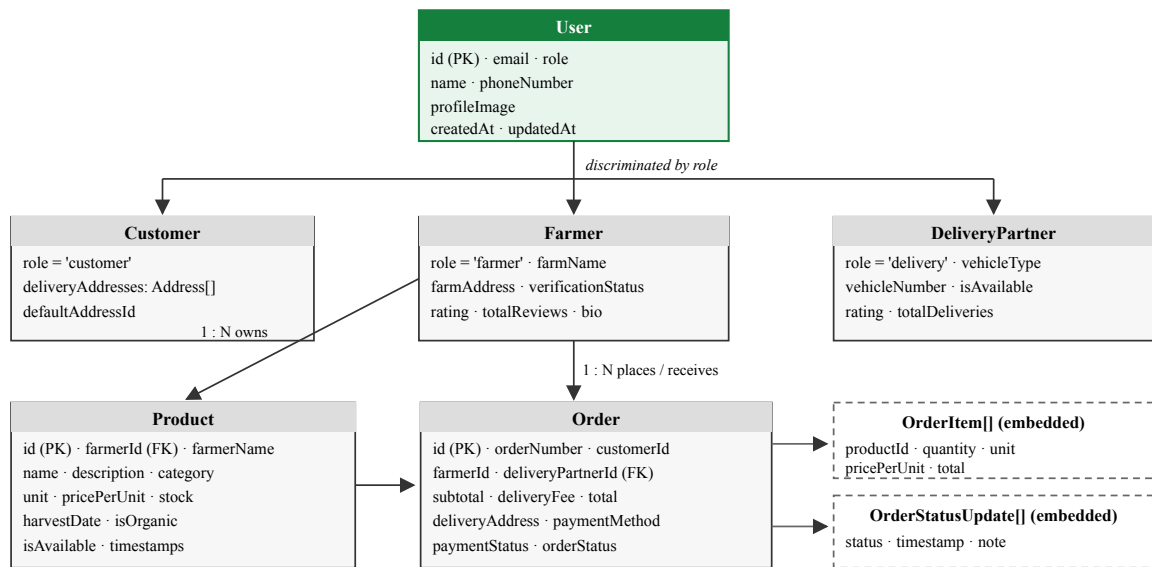


Figure 2.4 — Component interaction sequence for order placement, including the per-farmer split and the stock decrement.

2.1.4 Data Model

The domain is expressed as TypeScript interfaces in `src/types/index.ts` and mirrors the relational schema used when Supabase is the active target. `User` is the base identity; `Customer`, `Farmer` and `DeliveryPartner` extend it with a discriminated `role` literal, which lets the compiler narrow a value to the correct specialisation from the `role` field alone. Orders embed their line items and their status history as nested structures rather than referencing separate tables, so that an order is retrievable in a single read and its history cannot drift from the order it describes.



Review and Subscription are defined in the type system; Review has service methods, Subscription is reserved for future work.

Figure 2.5 — Entity relationship model of the persisted domain. Dashed boxes are embedded (denormalised) structures rather than separate tables.

Entity	Principal attributes	Role that owns writes
User	id, email, role, name, phoneNumber, timestamps	Self (profile edit); role set once at sign-in
Farmer	farmName, farmAddress, verificationStatus, rating, totalReviews, bio	Farmer; verification reserved for an administrator
Product	farmerId, name, category, unit, pricePerUnit, stock, harvestDate, isOrganic, isAvailable	Farmer (create, price, toggle visibility); system decrements stock on order
Order	orderNumber, customerId, farmerId, deliveryPartnerId, items[], subtotal, deliveryFee, total, deliveryAddress, paymentMethod, paymentStatus, orderStatus, statusHistory[]	Customer creates and may cancel; farmer advances to <i>packed</i> ; delivery partner advances to <i>delivered</i>
Address	label, street, city, state, pincode, latitude, longitude, isDefault	Customer at checkout; farmer for farm location
Review	orderId, customerId, farmerId, rating, comment, images	Customer — service layer implemented, authoring UI deferred

Table 2.3 — Domain entities and principal attributes.

2.1.5 Navigation and Route Guarding

Navigation is declarative. Each role directory contains a `_layout.tsx` that declares a `Tabs` navigator with a shared custom tab bar; screens that must be reachable but not tabbed — product detail, order tracking and the payment screen — are declared with `href: null` so they participate in the stack without occupying a tab. Above all three groups, the root layout runs a guard on every segment change. Figure 2.6 shows the resulting graph and the three redirect rules the guard enforces.

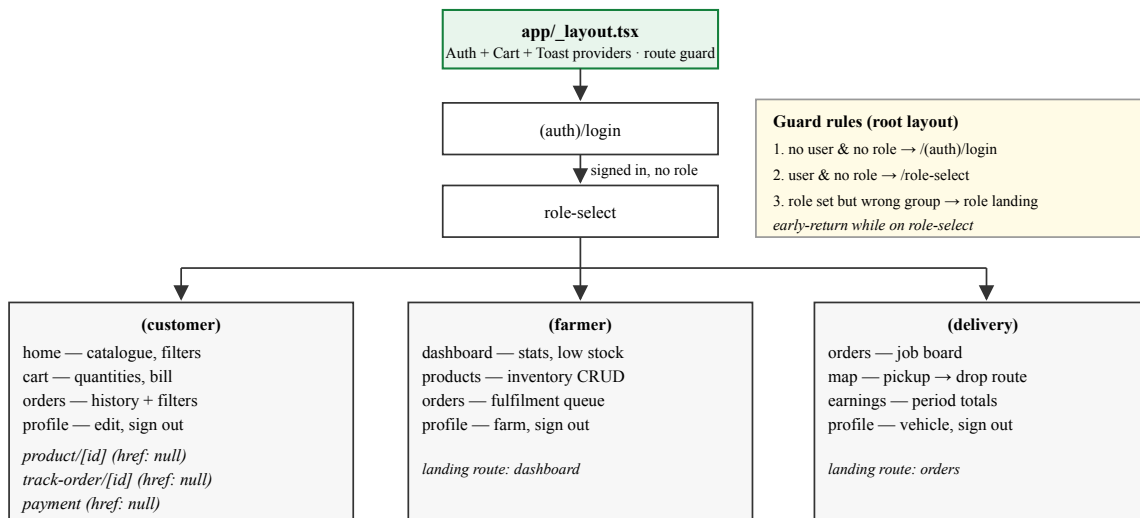


Figure 2.6 — Navigation map of routed screens and the redirect rules enforced by the root guard.

2.2 Technology Stack

2.2.1 Programming Languages

The application is written entirely in **TypeScript 5.9** with `strict` enabled in `tsconfig.json`. Strict mode is not decoration here: the domain model leans on discriminated unions — `UserRole`, `OrderStatus`, `ProductCategory` are string-literal unions rather than free-form strings — so the compiler rejects an invalid status transition or an unhandled category at build time rather than at run time. **JSX/TSX** is used for all component markup. A small amount of plain **JavaScript** remains in the Babel, Metro and Tailwind configuration files and in the unused Express scaffold under `backend/`.

2.2.2 Frameworks and Libraries

Technology	Version	Role in the system and rationale for the choice
------------	---------	---

React Native	0.81.5	Cross-platform UI runtime. One codebase compiles to Android, iOS and — through react-native-web — the browser, which is what allows the whole system to be driven by the automated test suites in a desktop browser.
React	19.1.0	Component model, hooks and the Context API used for global state.
Expo SDK	54	Managed workflow. Supplies the development client, the Metro bundler configuration, over-the-air asset handling and native module access without ejecting to bare native projects.
Expo Router	6.0	File-based routing with typed routes enabled. The directory layout <i>is</i> the navigation graph, which keeps the three role workspaces physically separated in the source tree.
TypeScript	5.9	Static typing in strict mode across all 48 source files.
NativeWind + Tailwind CSS	4.0 / 3.4	Utility-class styling compiled to React Native styles, with a project palette (primary green, earth brown) extended in the Tailwind config.
Supabase JS	2.91	Client for the optional hosted backend: GoTrue authentication with JWT session persistence, PostgREST queries against PostgreSQL, and object storage for uploads.
AsyncStorage	2.2	Device key-value persistence for the session user, the selected role, the cart and the entire on-device data layer.
React Native Reanimated / Worklets	4.1 / 0.5	Native-thread animation driver underpinning gesture and transition work.
Moti	0.29	Declarative animation helper retained as a dependency; the shipped entrance animations use the React Native <code>Animated</code> driver instead, because Moti intermittently stalled at its initial state on web.
expo-linear-gradient	15.0	Brand gradient headers and primary buttons.
@expo/vector-icons	15.0	Ionicons glyph set used throughout the interface and in the tab bar.
expo-haptics	15.0	Selection feedback on press, suppressed on the web target.
expo-image-picker / document-picker	17.0 / 14.0	Media and document selection, wired for the deferred image-upload and verification features.
react-native-safe-area-context, screens, gesture-handler	5.6 / 4.16 / 2.28	Navigation primitives required by the Expo Router stack and tab navigators.

Table 2.1 — Technology stack with versions and rationale.

2.2.3 Tools and Platforms

- **Node.js 22.14 and npm 10.9** — package management and script execution.
- **Metro bundler** — JavaScript bundling for all three targets, configured through `metro.config.js` and wrapped with the NativeWind transformer.
- **Babel** with `babel-preset-expo`, the NativeWind JSX import source and the Reanimated worklets plugin.
- **Expo Go** — on-device execution of the development build on Android and iOS without a native toolchain.
- **Git and GitHub** — version control; forty-three commits over the project timeline.
- **Chrome and the Chrome DevTools Protocol** — the web target was driven programmatically through Puppeteer to execute the end-to-end test walkthrough described in Chapter 3.
- **Supabase** — optional managed PostgreSQL, authentication and storage platform.
- **Prettier** — source formatting, configured in `.prettierrc`.

2.3 System Modules

The system decomposes into six modules: one shared foundation, one authentication module and one module per role. Table 2.2 summarises them; the subsections that follow describe the functional flow of each.

Module	Source location	Responsibility
Foundation	<code>src/theme.ts</code> , <code>src/constants/</code> , <code>src/utils/</code> , <code>src/components/ui/</code>	Design tokens, business-rule constants, formatting and validation helpers, and twelve reusable presentational components.
Data access	<code>src/services/</code> , <code>src/config/</code>	Service façade, on-device data layer, device storage wrapper and Supabase client configuration.
Authentication	<code>app/(auth)/</code> , <code>app/role-select.tsx</code> , <code>src/contexts/AuthContext.tsx</code>	Sign-up, sign-in, role assignment, session restoration and the global navigation guard.
Customer	<code>app/(customer)/</code> , <code>src/contexts/CartContext.tsx</code>	Catalogue browsing, product detail, cart, checkout, order history and order tracking.
Farmer	<code>app/(farmer)/</code>	Business dashboard, inventory management and order fulfilment.

Delivery	app/(delivery) /	Availability, job board, route view and earnings.
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Table 2.2 — Module-wise responsibility summary.

2.3.1 Authentication and Role Assignment Module

A new user opens the application and is redirected to the login screen by the root guard. The sign-up form collects a name, an email, a ten-digit Indian mobile number and a password; each is validated client-side before any network call is attempted. On successful registration the user is deliberately signed out again and returned to the sign-in tab, so that the session that eventually drives the application is always a genuine sign-in session rather than an artefact of registration.

On sign-in, the service returns a user record. If that record already carries a role, the user is routed directly to the corresponding workspace. If it does not, the user is sent to the role selection screen, where choosing *Farmer* additionally provisions a farmer profile with a derived farm name and a *pending* verification status. The chosen role is written both to the user record and to device storage, so that a subsequent cold start restores the workspace without a round trip.

The guard in `app/_layout.tsx` re-evaluates on every navigation segment change and enforces three rules: an unauthenticated user with no stored role is sent to login; an authenticated user without a role is sent to role selection; and a user with a role who is found anywhere outside their own route group is redirected to that group's landing screen. The guard deliberately early-returns while the user is on the role selection screen, which prevents the redirect loop that would otherwise occur during the brief window in which a user exists but a role does not.

2.3.2 Customer Module

The customer home screen loads the available catalogue and the verified farmer list in parallel. Products are presented as a two-column card grid; each card shows the category glyph, an organic badge where applicable, the farmer's name, the unit price and a one-tap add control. Above the grid the customer can search by name or description, filter by any of the eight categories, restrict the list to organic produce, and sort by ascending or descending price. Filtering by category re-queries the data source; search, organic filtering and sorting are applied client-side to the loaded set, which keeps typing responsive.

Selecting a card opens the product detail screen, which adds the harvest provenance, a stock badge that switches to an out-of-stock state at zero, a quantity stepper clamped to available stock and a farmer attribution card. Adding to the cart from either surface writes through

`CartContext`, which merges quantities for a product already in the cart and caps the merged quantity at the product's stock.

The cart screen renders each line with a stepper and a remove control, and adds a progress indicator toward the free-delivery threshold that switches to a celebratory state once the threshold is crossed. The bill summary computes subtotal, delivery fee — zero above the threshold — and total. Checkout is blocked below the minimum order value, with the shortfall stated explicitly in the error message.

The checkout screen collects a delivery address, validates the pincode against the Indian six-digit pattern, offers cash on delivery, UPI and card, and then places the order. Because a cart may contain produce from several farmers, the cart is partitioned by farmer and one order is created per farmer, each with its own subtotal and its own fulfilment lifecycle. Payment status is recorded as *pending* for cash on delivery and *paid* for the two prepaid methods.

Order history lists all of the customer's orders newest first, filterable by all, active, delivered or cancelled. Each entry opens a tracking screen that renders the seven-stage timeline with completed stages marked, the current stage annotated *in progress*, an estimated delivery time derived from the order timestamp, the itemised bill, the delivery address and — while the order is still in *placed* or *confirmed* — a cancellation control.

2.3.3 Farmer Module

The farmer dashboard aggregates the farmer's orders and products into four indicators: active orders, active products, lifetime order count and realised earnings, the last computed as the sum of the totals of delivered orders only. Beneath the indicators, any product whose stock has fallen to or below the low-stock threshold is surfaced in an amber alert list with a direct link to inventory, followed by the five most recent orders with their status pills.

The products screen is the farmer's inventory console. It lists every product the farmer owns — including those currently hidden from customers, so that they can be restored — with search across name and category, a low-stock marker and an availability switch that toggles catalogue visibility without deleting the record. A modal sheet creates a new product, collecting name, description, category, unit, price, stock and an organic flag, with name, price and stock enforced as mandatory.

The orders screen is the fulfilment queue, filterable by active, all or completed. The farmer's authority over an order is deliberately bounded: the transition map allows only *placed* → *confirmed* and *confirmed* → *packed*. Once an order is packed it leaves the farmer's control and becomes visible to delivery partners. Every transition is confirmed through a dialogue before it is committed.

2.3.4 Delivery Module

The delivery partner's job board opens with an online/offline switch and two live counters for active deliveries and the fee accrued against them. When online, the partner sees every packed order that has not yet been claimed, each showing the drop address and — where the address carries coordinates — a straight-line distance and an estimated duration. Declining a job hides it locally for the remainder of the session; accepting it atomically sets the order status to *assigned* and stamps the partner's identifier onto the order, which removes it from every other partner's board.

Claimed orders move into *My Deliveries*, where a single contextual action advances the order along the remainder of the lifecycle: *Pick Up* at *assigned*, *Start Delivery* at *picked up*, and *Delivered* at *out for delivery*. Marking an order delivered also flips its payment status to *paid*, which closes the cash-on-delivery loop.

The route screen resolves the partner's single in-flight delivery and presents it as a pickup-to-drop schematic with the farm as origin and the customer address as destination, together with the order value and the same contextual advance action. The earnings screen aggregates all delivered orders into today, week-to-date and month-to-date totals at the standard per-delivery fee, and lists the ten most recent completed deliveries with relative timestamps.

2.4 Key Algorithms and Logic

Six pieces of logic carry most of the system's behaviour. Each is presented below as pseudocode followed by the reasoning behind it and, where instructive, the actual implementation.

2.4.1 Role-Based Navigation Guard

The guard is the single point at which authorisation is decided. Because Expo Router exposes the current URL as a segment array, the guard can compare the user's role against the segment they are attempting to occupy without any per-screen checks.


```

ALGORITHM EnforceRouteAccess
INPUT      user, role, segments      // segments[0] is the active route group
OUTPUT     a redirect, or no action

1  IF authentication is still loading THEN return
2  group ← segments[0]
3  IF group = 'role-select' THEN return          // avoid redirect loop
4  IF role is set AND group = groupFor(role) THEN return // already correct
5  IF user is null AND role is null AND group ≠ '(auth)' THEN
6      redirect '/(auth)/login'
7  ELSE IF (user OR role) AND role is null THEN
8      redirect '/role-select'
9  ELSE IF role is set THEN
10     IF group ≠ groupFor(role) OR group ∈ {'(auth)', 'role-select'} THEN
11         redirect landingFor(role)
        // groupFor : customer→(customer), farmer→(farmer), delivery→(delivery)
        // landingFor: customer→/home, farmer→/dashboard, delivery→/orders

```

Listing 2.1 — Route access enforcement (*app/_layout.tsx*).

Line 3 is the non-obvious one. During role assignment there is a window in which a user exists but a role does not; without the early return, rule 7 would fire on every render and pin the user to the role selection screen while the assignment was still in flight. Line 4 short-circuits the common case so that ordinary tab navigation inside a workspace performs no redirect work at all.

2.4.2 Order Lifecycle State Machine

An order occupies one of eight states. Seven of them form a strictly linear progression; *cancelled* is a terminal state reachable only from the two earliest states. What makes the machine safe is that no single role can drive an order from end to end: authority is partitioned, so a farmer cannot mark an order delivered and a delivery partner cannot confirm one.

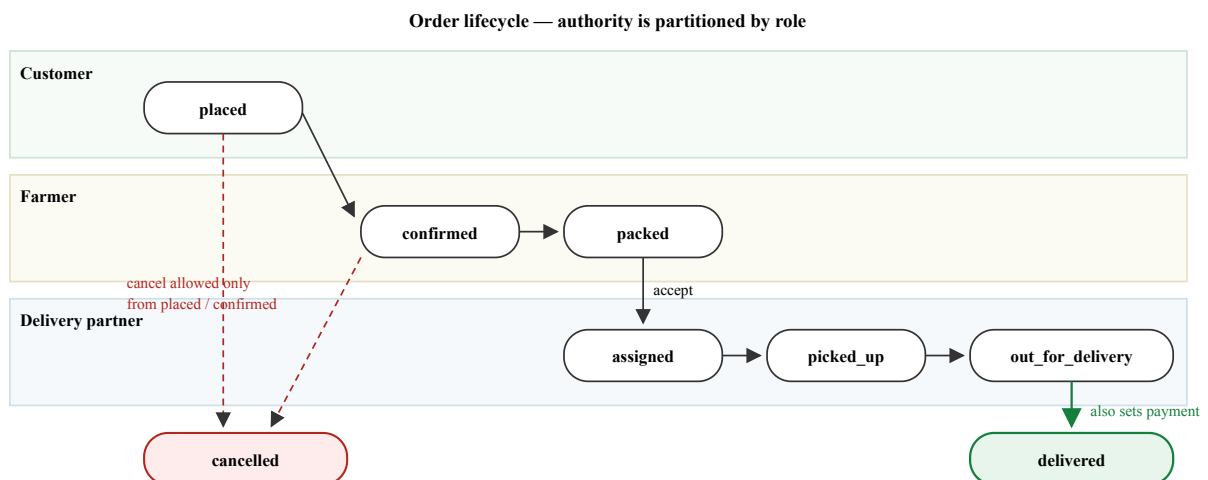


Figure 2.7 — Order lifecycle state machine. Each swim lane shows which role is permitted to originate the transitions drawn within it.

```

ALGORITHM  AdvanceOrderStatus
INPUT      orderId, nextStatus, actorRole, note?
OUTPUT     persisted order with an appended history entry

1  order ← fetch(orderId)
2  IF order is null THEN return
3  allowed ← transitionsPermittedFor(actorRole, order.orderStatus)
4  IF nextStatus ∉ allowed THEN reject
5  order.orderStatus ← nextStatus
6  order.statusHistory ← order.statusHistory + { nextStatus, now(), note }
7  IF nextStatus = 'delivered' THEN order.paymentStatus ← 'paid'
8  order.updatedAt ← now()
9  persist(order)

transitionsPermittedFor(role, current):
  farmer   : placed→confirmed, confirmed→packed
  delivery : packed→assigned, assigned→picked_up,
               picked_up→out_for_delivery, out_for_delivery→delivered
  customer : placed→cancelled, confirmed→cancelled

```

Listing 2.2 — Guarded status transition with history append.

The history is append-only. Because each entry carries its own timestamp, the tracking timeline can be rendered purely from history without inferring times, and an audit of how long an order spent in each stage remains possible after the fact. In the implementation each role's screen owns its own transition table — `nextStatusMap` on the farmer's order queue and `statusMap` on the delivery job board — which is what enforces line 4 in practice.

2.4.3 Cart Quantity Reconciliation and Stock Capping

The cart must never promise more of a product than exists. Two entry points can violate that — adding a product already present, and typing a quantity directly — so the cap is applied inside the reducer rather than at either call site.

```

ALGORITHM  AddToCart
INPUT      product, quantity
OUTPUT     next cart state

1  cap(q) ← IF product.stock > 0 THEN min(q, product.stock) ELSE q
2  existing ← cart.find(line ⇒ line.productId = product.id)
3  IF existing exists THEN
4    RETURN cart with existing.quantity replaced by cap(existing.quantity +
    quantity)
5  ELSE
6    RETURN cart + { productId, product, quantity: cap(quantity), farmerId }

ALGORITHM  UpdateQuantity(productId, q)
1  IF q ≤ 0 THEN RETURN RemoveFromCart(productId)
2  maxStock ← line.product.stock > 0 ? line.product.stock : q
3  line.quantity ← min(q, maxStock)

```

Listing 2.3 — Cart reconciliation with stock capping (CartContext.tsx).

Persistence introduces a subtler hazard. The cart hydrates from storage asynchronously, while the effect that writes the cart back fires on the very first render — when the cart is still the empty initial state. Writing at that moment would erase the stored cart before it had been read. The implementation guards the write with a hydration flag that is only raised once the initial read has settled:

```
const hydratedRef = useRef(false);

useEffect(() => { loadCart(); }, []);

useEffect(() => {
  // Don't overwrite the persisted cart with [] before loadCart resolves
  if (hydratedRef.current) saveCart();
}, [cart]);

const loadCart = async () => {
  try {
    const data = await AsyncStorage.getItem(CART_KEY);
    if (data) setCart(JSON.parse(data));
  } catch (error) {
    console.error('Error loading cart:', error);
  } finally {
    hydratedRef.current = true;
  }
};
```

Listing 2.4 — Hydration-guarded cart persistence (src/contexts/CartContext.tsx).

2.4.4 Order Splitting, Pricing and Fee Rules

A cart is a customer-side concept; an order is a farmer-side concept. Since one farmer cannot fulfil another's produce, a mixed cart must become several orders. The delivery fee, however, is charged once against the cart as a whole and is waived entirely above the free-delivery threshold.

```

ALGORITHM PlaceOrder
INPUT      cart, address, paymentMethod, user
OUTPUT     one persisted order per distinct farmer

1  IF user is null          THEN prompt sign-in;      abort
2  IF cart is empty        THEN report;              abort
3  IF cartTotal < MIN_ORDER_VALUE THEN report shortfall; abort
4  IF any address field is blank THEN report;        abort
5  IF NOT matches(address.pincodes, /^[1-9][0-9]{5}$/) THEN report; abort
6  deliveryFee ← cartTotal ≥ FREE_DELIVERY_THRESHOLD ? 0 : DELIVERY_FEE
7  groups ← groupBy(cart, line ⇒ line.farmerId)
8  FOR EACH (farmerId, lines) IN groups IN PARALLEL
9      subtotal ← Σ line.product.pricePerUnit × line.quantity
10     createOrder({ orderNumber: unique(), customerId, farmerId,
                     items: map(lines), subtotal, deliveryFee,
                     total: subtotal + deliveryFee, deliveryAddress: address,
                     paymentMethod,
                     paymentStatus: paymentMethod = 'cod' ? 'pending' : 'paid',
                     orderStatus: 'placed',
                     statusHistory: [{ status:'placed', timestamp: now() }] })

11 AWAIT all
12 clearCart(); redirect to order history

```

Listing 2.5 — Checkout, validation and per-farmer order splitting (app/(customer)/payment.tsx).

Every threshold referenced above is a named constant in a single module rather than a literal scattered through the screens, so a change to the delivery fee or the free-delivery threshold takes effect consistently in the cart, the checkout screen, the delivery job board and the earnings calculation at once.

Constant	Value	Effect on system behaviour
MIN_ORDER_VALUE	₹100	Checkout is refused below this subtotal; the shortfall is reported to the customer.
DELIVERY_FEE	₹30	Charged once per order; also the per-delivery remuneration credited to the partner.
FREE_DELIVERY_THRESHOLD	₹500	Above this subtotal the fee is waived; drives the cart progress indicator.
LOW_STOCK_THRESHOLD	10 units	Triggers the farmer dashboard alert and the inventory warning marker.
MAX_DISTANCE_KM	50 km	Radius used when querying nearby farmers.
ESTIMATED_DELIVERY_MINUTES	45 min	Baseline promise used for the customer's estimated delivery time.
MINUTES_PER_KM	4 min/km	Distance-to-duration factor for the delivery partner's ETA.

REGEX.phone	<code>/^[6-9]\d{9}\$/</code>	Indian mobile format enforced at sign-up and on profile edit.
REGEX.pincode	<code>/^[1-9][0-9]{5}\$/</code>	Six-digit postal code enforced at checkout.

Table 2.4 — Centralised business rule constants (*src/constants/index.ts*).

2.4.5 Distance and Estimated Time of Arrival

The delivery job board ranks and annotates jobs by how far away the drop address is. Distance is computed with the Haversine formula over the great-circle path, using the partner's origin (a fixed reference point in the current build, since live location is out of scope) and the order's stored coordinates. The resulting distance is converted to a duration and floored at the baseline promise, so an ETA is never optimistic relative to the figure already shown to the customer.

```

ALGORITHM DistanceAndETA
INPUT      (lat1, lon1) origin, (lat2, lon2) destination
OUTPUT     distance in km, estimated minutes

1  R ← 6371 // mean Earth radius, km
2  Δφ ← radians(lat2 - lat1); Δλ ← radians(lon2 - lon1)
3  a ← sin2(Δφ/2) + cos(radians(lat1)) · cos(radians(lat2)) · sin2(Δλ/2)
4  c ← 2 · atan2(√a, √(1-a))
5  d ← R · c
6  IF d is not finite OR d ≤ 0 THEN RETURN (d, ESTIMATED_DELIVERY_MINUTES)
7  RETURN (d, max(ESTIMATED_DELIVERY_MINUTES, round(d × MINUTES_PER_KM)))

```

Listing 2.6 — Haversine distance and ETA derivation (*src/utils/helpers.ts*).

2.4.6 Dual-Target Persistence Dispatch

The mechanism that makes the backend substitutable is deliberately unremarkable: a boolean computed once at module load, and an early return at the head of every service method. The boolean is true only when an explicit Supabase URL has been supplied and it is not the placeholder value, so a developer who checks out the repository and runs it with no configuration is served by the on-device data layer rather than by a broken remote.

```

export const isSupabaseConfigured = Boolean(
  process.env.EXPO_PUBLIC_SUPABASE_URL &&
  process.env.EXPO_PUBLIC_SUPABASE_URL !== 'https://placeholder.supabase.co'
);

static async getAllProducts(limit: number = 50): Promise<Product[]> {
  if (!isSupabaseConfigured) {
    return localStore.getAvailableProducts(); // ← local branch
  }
  const { data, error } = await supabase // ← remote branch
    .from('products')
    .select('*')
    .eq('is_available', true)
    .order('created_at', { ascending: false })
    .limit(limit);
  if (error) throw error;
  return (data || []).map(item => ({ // snake_case → camelCase
    id: item.id, farmerId: item.farmer_id, farmerName: item.farmer_name,
    name: item.name, category: item.category, unit: item.unit,
    pricePerUnit: item.price_per_unit, stock: item.stock,
    harvestDate: new Date(item.harvest_date), isOrganic: item.is_organic,
    isAvailable: item.is_available, /* ... */
  }));
}

```

Listing 2.7 — Dual-target dispatch in the service façade (src/services/supabase.ts).

The on-device layer is not a stub that returns fixed arrays. It maintains genuine mutable state — creating an order decrements the stock of every product in it, advancing a status appends to that order's history, and adding a product makes it appear in the customer catalogue — and mirrors that state to AsyncStorage after every mutation. Rehydration uses a JSON reviver keyed on the known date fields, so `Date` instances survive the round trip and date arithmetic in the UI does not silently receive strings:

```

const DATE_KEYS = new Set(['createdAt', 'updatedAt', 'harvestDate', 'timestamp',
                           'nextDeliveryDate', 'startDate']);

function reviveDates(key: string, value: unknown) {
  return DATE_KEYS.has(key) && typeof value === 'string' ? new Date(value) :
  value;
}

async function ensureLoaded(): Promise<LocalState> {
  if (state) return state;
  if (!loadPromise) {
    loadPromise = (async () => {
      try {
        const raw = await AsyncStorage.getItem(STORE_KEY);
        if (raw) { state = JSON.parse(raw, reviveDates); return state!; }
      } catch (error) { console.error('localStorage: failed to load, restoring
defaults', error); }
      state = { products: [...PRODUCTS], orders: [...ORDERS], reviews: [] };
      return state;
    })();
  }
  return loadPromise; // single in-flight promise – concurrent callers share
it
}

```

Listing 2.8— On-device store hydration with date revival (src/services/localStore.ts).

Service group	Ops	Remote branch (Supabase)	Local branch (on-device store)
Authentication	6	GoTrue sign-up, sign-in, sign-out, password reset, session read, auth state subscription	Deterministic identifiers derived from the email; a null session and a no-op subscription
Users	3	users table insert, select by id, update with camel→snake mapping	Read/merge/write against the AsyncStorage user record
Farmers	3	farmers table insert, select, and verified-farmer listing	Verified farmer profiles held on device
Products	6	Insert, select by id, by farmer, all available, by category, update, delete	Filtered reads and mutations against the in-memory product array
Orders	8	Insert, select by id, by customer, by farmer, by partner, delivered-by-partner, update, status update	Equivalent reads with stock decrement on create and history append on status change

Reviews	3	Insert, list by farmer, recompute farmer average rating	Insert and list against the in-memory review array
Storage	1	Upload to the <code>uploads</code> bucket, return the public URL	Return the local device URI unchanged

Table 2.5 — Service façade operations and dual-target dispatch.

2.5 Screenshots and Code Snippets

Every screenshot in this section was captured from the running application. The web target was launched on the local Metro development server and driven programmatically through the Chrome DevTools Protocol at a viewport of 430×932 logical pixels at $2\times$ device pixel ratio, which corresponds to a contemporary handset. The captures form a single continuous session: the order created by the customer in Figure 2.10 is the same order confirmed and packed by the farmer in Figure 2.13 and subsequently claimed by the delivery partner in Figure 2.14. No screen is a mock-up and no data was inserted by hand.

2.5.1 Authentication and Role Assignment

The figure displays three mobile app screens for 'Farmer Groceries'. The first screen, (a) Sign-in, features a green header with the app logo and name, a segmented control for 'Sign In' and 'Sign Up', a 'Welcome back' message, and input fields for email and password. The second screen, (b) Sign-up with name and mobile, has a similar header and segmented control, but includes fields for full name, email, phone number, and password. The third screen, (c) Role selection, shows three role options: Customer, Farmer, and Delivery Partner, each with a description and a radio button. A 'Continue' button is at the bottom. The fourth screen, (d) Customer role selected, shows the 'Customer' role selected with a green checkmark, while the other roles remain unselected.

(a) Sign-in

(b) Sign-up with name and mobile

(c) Role selection

(d) Customer role selected

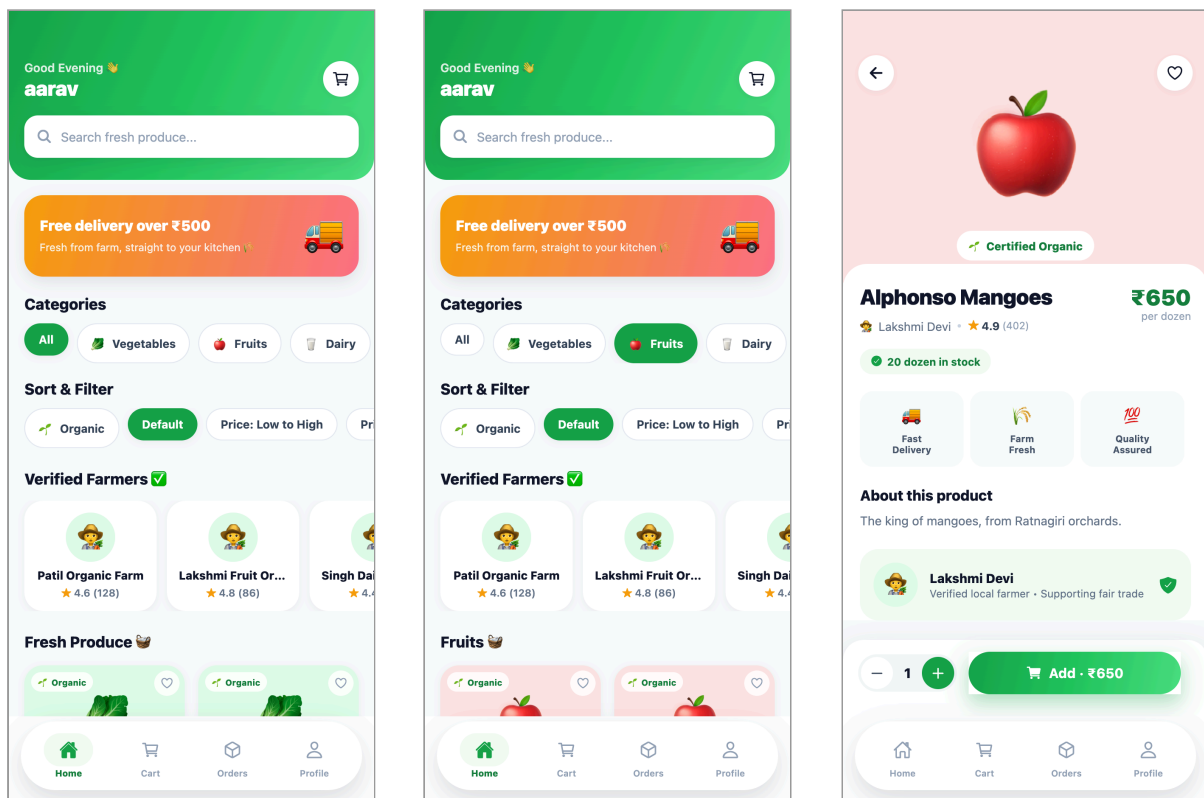
Figure 2.8 — Authentication and role selection. The segmented control switches between sign-in and sign-up in place; role selection is presented once and then persisted.

The login screen validates before it calls the network. Email format, password length and — for sign-up — name and a ten-digit mobile number are all checked locally, and each failure surfaces as a toast rather than an inline field error, which keeps the form compact on a small viewport:

```
const handleAuth = async () => {
  if (!email || !password) return toast.show('Please enter email and password',
    'error');
  if (isSignUp && !name) return toast.show('Please enter your name',
    'error');
  if (isSignUp && !phone) return toast.show('Please enter your phone number',
    'error');
  if (!validateEmail(email.trim()))
    return toast.show('Please enter a valid email address', 'error');
  if (password.length < 6)
    return toast.show('Password must be at least 6 characters', 'error');
  if (isSignUp && !validatePhone(phone.replace(/\D/g, '').slice(-10)))
    return toast.show('Enter a valid 10-digit mobile number', 'error');
  /* ... registration or sign-in ... */
};
```

Listing 2.9 — Client-side credential validation (*app/(auth)/login.tsx*).

2.5.2 Customer Module



(a) Catalogue with categories, sort and filter controls, and verified farmers

(b) Catalogue filtered to the Fruits category

(c) Product detail with stock badge, quantity stepper and farmer attribution

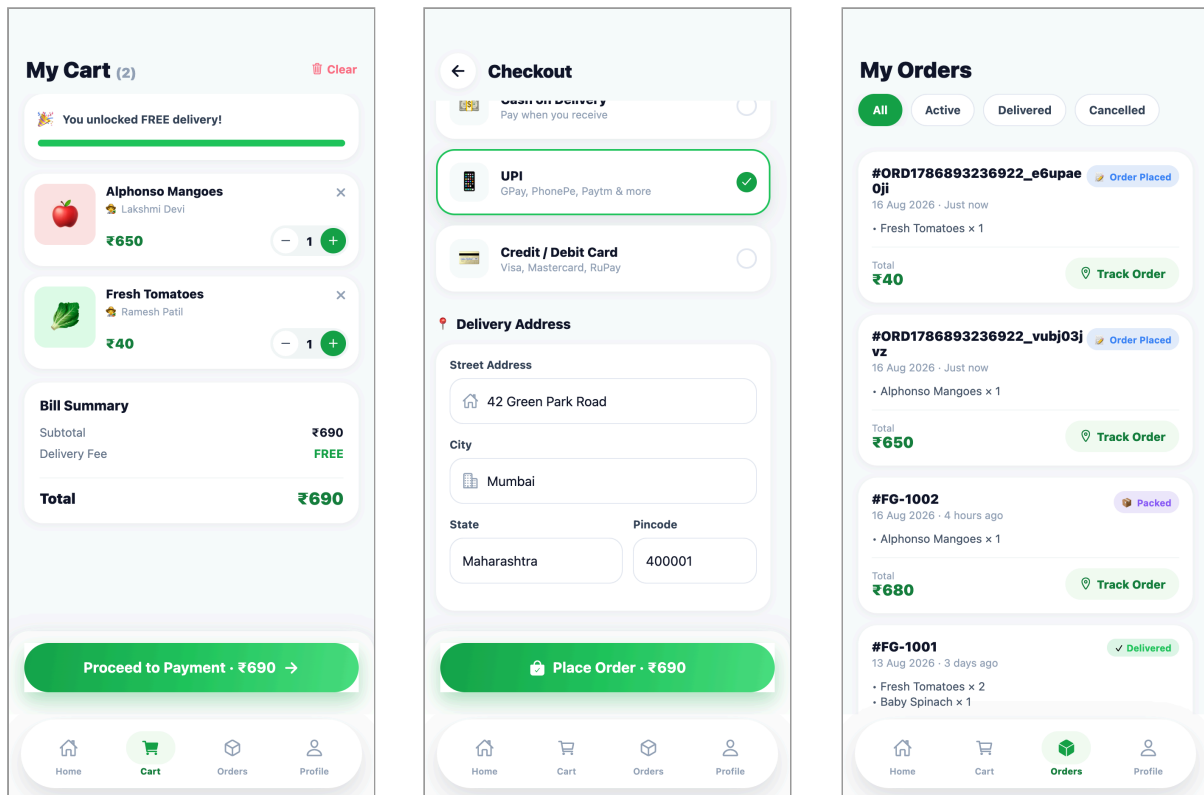
Figure 2.9 — Customer module: catalogue discovery, filtering and product detail.

Search, organic filtering and price sorting are composed as a single derived list rather than three separate passes over state, so the rendered grid is always a pure function of the loaded

products and the four control values:

```
const filteredProducts = products
  .filter((p) => p.name.toLowerCase().includes(searchQuery.toLowerCase())
    || (p.description ??
    '').toLowerCase().includes(searchQuery.toLowerCase()))
  .filter((p) => !organicOnly || p.isOrganic)
  .sort((a, b) => (sortBy === 'price_asc' ? a.pricePerUnit - b.pricePerUnit
    : sortBy === 'price_desc' ? b.pricePerUnit - a.pricePerUnit
    : 0));
```

Listing 2.10 — Composed catalogue filtering and sorting (*app/(customer)/home.tsx*).

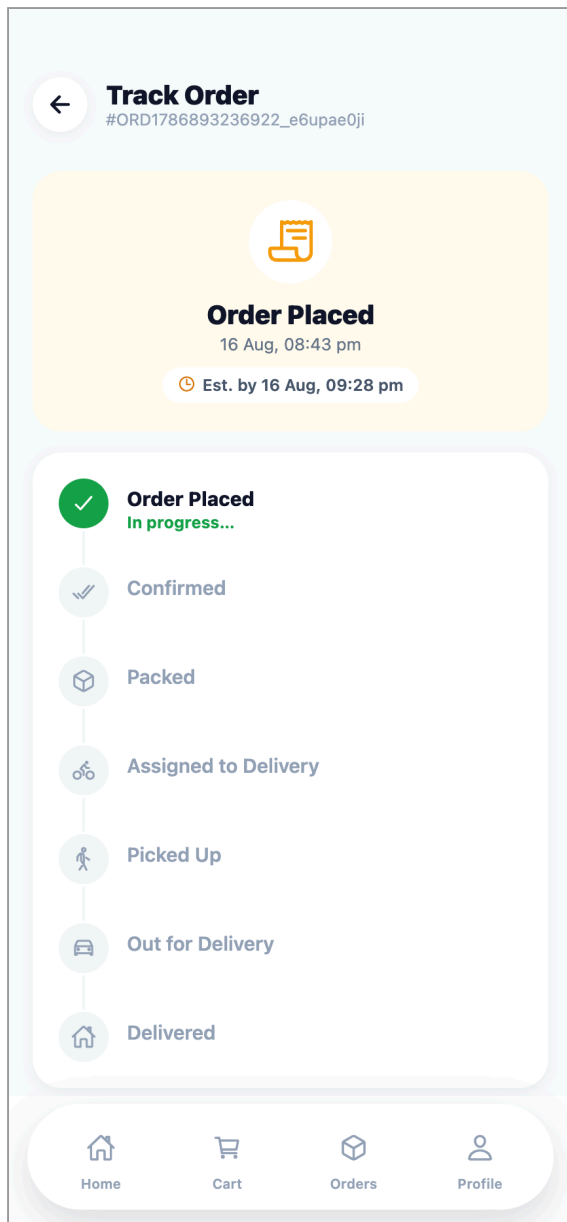


(a) Cart — free-delivery progress met, per-line steppers, bill summary

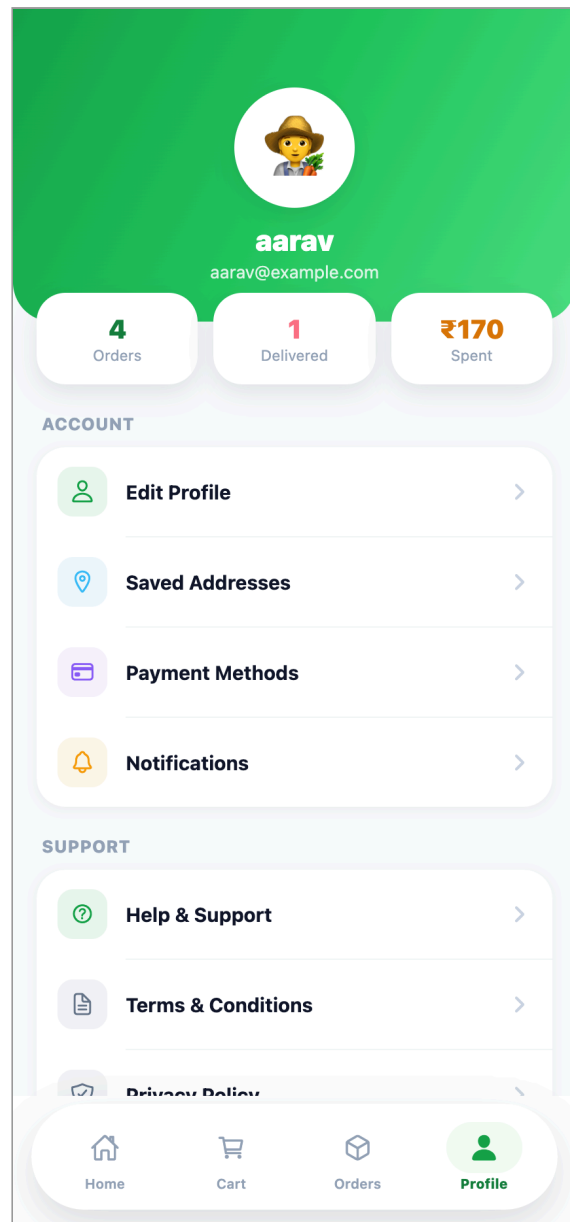
(b) Checkout — payment method selection and validated address entry

(c) Order history immediately after placement, showing the per-farmer split

Figure 2.10 — Customer module: cart, checkout and order history. The cart in (a) contains produce from two different farmers; (c) shows the two resulting orders.



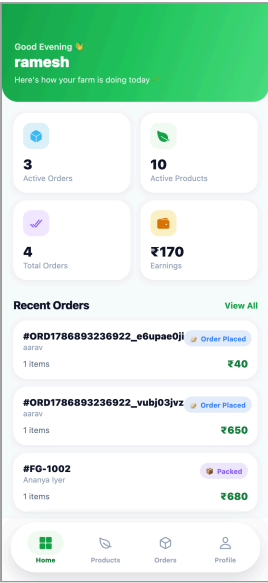
(a) Seven-stage tracking timeline with the current stage annotated and an estimated delivery time



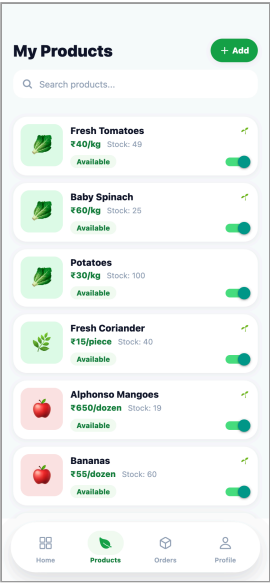
(b) Customer profile with editable details, grouped account and support actions

Figure 2.11 — Customer module: order tracking and profile management.

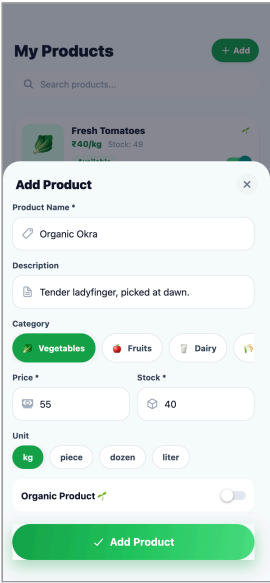
2.5.3 Farmer Module



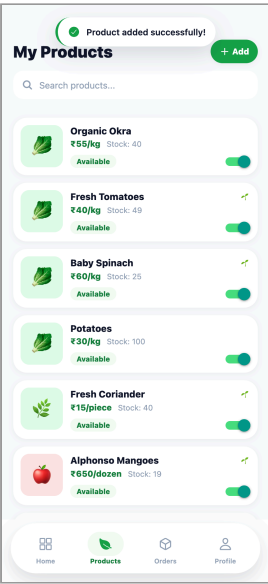
(a) Dashboard — four indicators and recent orders



(b) Inventory with search and visibility switches

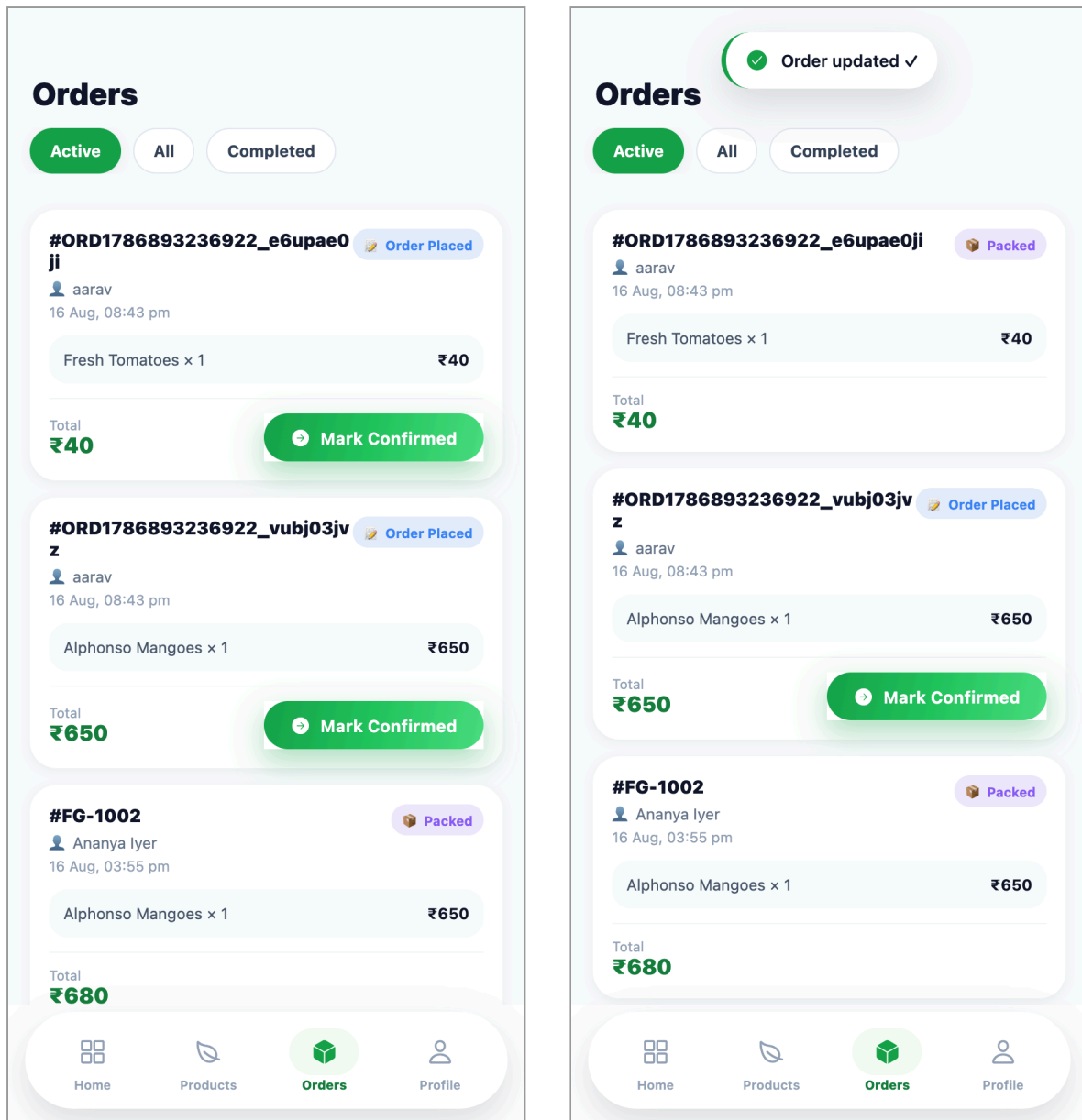


(c) Product creation sheet



(d) Inventory after the new product is saved

Figure 2.12 — Farmer module: business dashboard, inventory management and product creation. The dashboard in (a) reflects the two orders placed by the customer in Figure 2.10.



(a) Fulfilment queue filtered to active orders, each offering only its permitted next transition

(b) After confirming and packing, the order shows the Packed pill and offers no further farmer action

Figure 2.13 — Farmer module: order fulfilment and the boundary of farmer authority.

The bound on farmer authority is expressed as data, not as branching. A partial record maps only the two transitions a farmer may originate; any status absent from the map yields no action button at all, so the interface cannot offer an illegal transition:

```

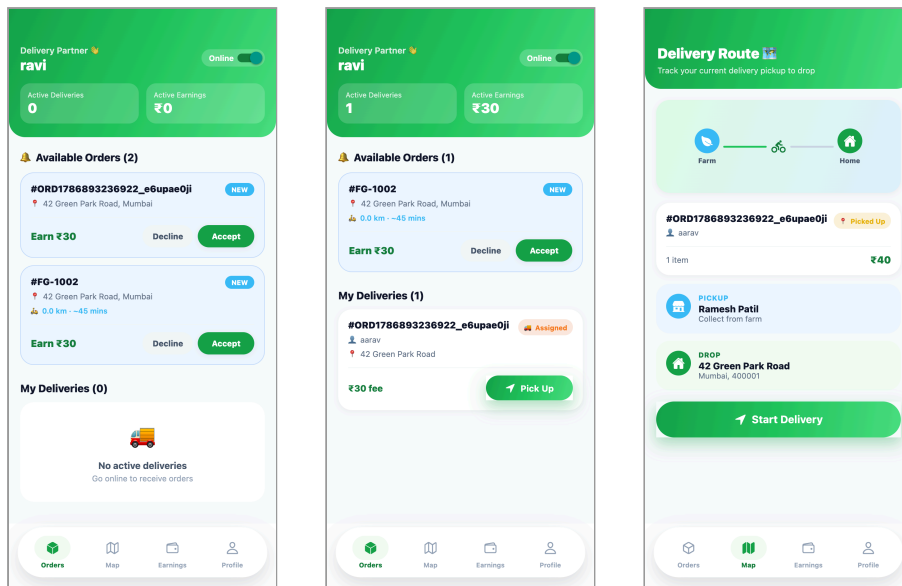
const nextStatusMap: Partial<Record<OrderStatus, OrderStatus>> = {
  placed: 'confirmed',
  confirmed: 'packed',
};

const nextStatus = nextStatusMap[order.orderStatus];
/* ... */
{nextStatus ? (
  <Button label={`Mark ${ORDER_STATUS[nextStatus]?.label}`} size="sm"
    icon="arrow-forward-circle"
    onPress={() => handleUpdateStatus(order.id, order.orderStatus)} />
) : null}

```

Listing 2.11 — Transition table restricting farmer authority (app/(farmer)/orders.tsx).

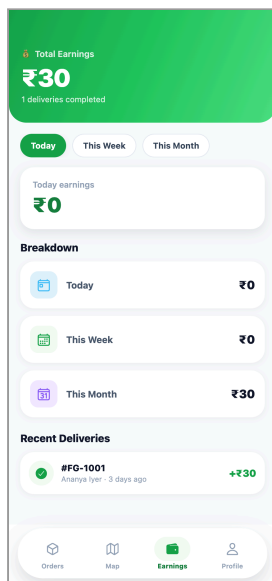
2.5.4 Delivery Module



(a) Job board — unclaimed packed orders with distance and ETA

(b) After acceptance the job moves to My Deliveries

(c) Pickup-to-drop route for the in-flight delivery



(d) Period-wise earnings and completed deliveries

Figure 2.14 — Delivery module: job board, claimed deliveries, route view and earnings. The first job in (a) is the order packed by the farmer in Figure 2.13.

Acceptance is the moment at which a job becomes exclusive to one partner. The status transition and the partner stamp are applied together, after which the order no longer satisfies the unclaimed-job predicate on any other partner's board:


```

const handleAcceptOrder = (orderId: string) => {
  Alert.alert('Accept Order', 'Accept this delivery?', [
    { text: 'Cancel', style: 'cancel' },
    { text: 'Accept', onPress: async () => {
      await SupabaseService.updateOrderStatus(orderId, 'assigned');
      await SupabaseService.updateOrder(orderId, { deliveryPartnerId:
user?.id });
      loadOrders();
      toast.show('Order accepted 🚚', 'success');
    } },
  ]);
};

// the board only offers jobs that are packed, unclaimed and not locally
declined
const availableOrders = isAvailable
  ? orders.filter((o) => !o.deliveryPartnerId
    && o.orderStatus === 'packed'
    && !declinedOrderIds.has(o.id))
  : [];

```

Listing 2.12 — Job acceptance and the unclaimed-job predicate (app/(delivery)/orders.tsx).

Earnings are derived rather than stored. A single pass over the partner's delivered orders accumulates four running totals against period boundaries computed once, which avoids three separate filtered traversals of the same array:

```

const stats = deliveredOrders.reduce((acc, order) => {
  const c = new Date(order.createdAt);
  acc.total += DELIVERY_FEE;
  if (c >= startOfMonth) acc.month += DELIVERY_FEE;
  if (c >= startOfWeek) acc.week += DELIVERY_FEE;
  if (c >= startOfDay) acc.today += DELIVERY_FEE;
  return acc;
}, { today: 0, week: 0, month: 0, total: 0 });

```

Listing 2.13 — Single-pass earnings aggregation (app/(delivery)/earnings.tsx).

2.5.5 Design System

Consistency across eighteen screens is maintained by a single token module. Palette entries, gradients, corner radii and a cross-platform elevation helper are declared once as plain values, so they can be consumed identically by NativeWind utility classes and by inline styles. The elevation helper is the only place in the codebase that has to know that web renders shadows as `box-shadow` while native platforms use shadow offsets and Android elevation:

```

const shadow = (y: number, blur: number, opacity: number, radius = 0):
ViewStyle =>
  Platform.select({
    web: { boxShadow: `0px ${y}px ${blur}px rgba(15, 23, 42, ${opacity})` },
    default: {
      shadowColor: '#0f172a',
      shadowOffset: { width: 0, height: y },
      shadowOpacity: opacity,
      shadowRadius: radius || blur / 2,
      elevation: y + 2,
    },
  }) as ViewStyle;

export const shadows = { sm: shadow(2, 8, 0.06), md: shadow(6, 18, 0.10), lg:
shadow(12, 30, 0.14) };
export const radii   = { sm: 10, md: 16, lg: 22, xl: 28, pill: 999 };

```

Listing 2.14 — Cross-platform elevation tokens (src/theme.ts).

CHAPTER 3: TESTING, VALIDATION AND RESULTS

3.1 Test Plan

3.1.1 Testing Strategy

The system under test is a client application whose correctness is dominated by two concerns: whether state changes made by one role become correctly visible to the others, and whether invalid input is refused before it reaches persistence. Neither concern is well served by unit tests of individual functions in isolation, because both are properties of a sequence of interactions across several screens. The strategy adopted therefore places its weight on **automated end-to-end testing against the running application**, supported by static verification.

Three complementary levels were applied:

1. **Static verification.** The entire source tree is compiled with `tsc --noEmit under strict`. Because the domain model uses string-literal unions for roles, order statuses and product categories, a large class of defects — an unhandled status, a mistyped category, a nullable value dereferenced without a guard — is rejected at this level before any test is run.
2. **Functional end-to-end testing.** The application is served on the web target and driven through a real Chrome instance over the DevTools Protocol. The suite, committed to the repository as `tests/e2e/walkthrough.mjs`, signs in as each of the three roles in turn and performs the complete business cycle: a customer places an order, the same order is confirmed and packed by the farmer, and the same order is then claimed and advanced by a delivery partner. This exercises cross-role propagation directly, because each role observes state written by the previous one.
3. **Validation and negative testing.** A second committed suite, `tests/e2e/validation.mjs`, deliberately supplies invalid input at every guarded boundary — malformed credentials, an unselected role, a sub-minimum cart, a blank address, an invalid pincode, a product created with mandatory fields empty — and asserts that the application refuses the operation and reports the specific reason. This suite also verifies two stateful properties that are easy to regress: that the cart survives an application restart, and that navigating directly to another role's URL is redirected.

Both suites exit with a non-zero status when any case fails, so they can be wired into a continuous integration pipeline unchanged, and both are runnable by any reader of the repository with `npm test` once `npm run web` is serving the application.

Interaction with the application is performed through DOM-level event dispatch rather than synthetic mouse input at coordinates. This was a necessary adaptation: react-native-web routes presses through its own responder system, which does not consistently observe input synthesised by the DevTools Protocol on every screen. Dispatching the full pointer and mouse event chain to the nearest pressable ancestor of a located text node reproduces a genuine user press reliably. Controlled text inputs are driven through the underlying value setter followed by an `input` event, which is the mechanism React itself observes.

3.1.2 Tools Used

Tool / component	Version	Purpose in the test process
TypeScript compiler	5.9	Static type verification of all 48 source files in strict mode.
Expo / Metro dev server	SDK 54	Serves the web target under test at <code>localhost:8081</code> .
Google Chrome (headless)	Current stable	Execution host for the application under test.
Puppeteer Core	23.x	Drives Chrome over the DevTools Protocol; performs presses, text entry, navigation, dialogue handling and screen capture.
<code>tests/e2e/walkthrough.mjs</code>	In repository	25-case functional scenario across all three roles; run with <code>npm run test:e2e</code> . Captures the 22 screenshots reproduced in §2.5 when <code>E2E_SHOTS=1</code> is set.
<code>tests/e2e/validation.mjs</code>	In repository	15 negative and stateful assertions; run with <code>npm run test:validation</code> .
<code>tests/e2e/helpers.mjs</code>	In repository	Shared press, text-entry, sign-in and assertion helpers used by both suites.
Device emulation	430 × 932 @ 2×	Handset-equivalent viewport for both capture and interaction.

Table 3.1 — Test environment and tooling.

3.2 Test Cases

Forty test cases were executed: twenty-five functional cases tracing the complete order lifecycle across the three roles, and fifteen validation cases probing the guarded boundaries. All inputs, expected outputs and observed statuses below are taken from the recorded execution logs of the two automated suites; no result is asserted from inspection of the source alone.

ID	Description	Input	Expected output	Status
TC-F01	Unauthenticated launch is redirected to sign-in	Cold start, empty device storage	Login screen is presented; no protected route is reachable	Pass
TC-F02	Sign-up mode reveals the additional identity fields	Press <i>Sign Up</i> on the segmented control	Full name and phone number fields appear; primary action becomes <i>Create Account</i>	Pass
TC-F03	Valid credentials sign the user in	aarav@example.com / test1234	Session established; user without a role is routed to role selection	Pass
TC-F04	Role selection routes to the correct workspace	Select <i>Customer</i> , press <i>Continue</i>	Role persisted; customer catalogue presented with the customer tab bar	Pass
TC-F05	Catalogue loads products and verified farmers	Customer home on first load	Ten products in a two-column grid; verified-farmer carousel with ratings	Pass
TC-F06	Category filter narrows the catalogue	Select the <i>Fruits</i> chip	Only fruit-category products remain; heading changes to <i>Fruits</i>	Pass
TC-F07	Product detail opens with stock and pricing	Open <i>Alphonso Mangoes</i>	Detail screen shows ₹650 per dozen, in-stock badge, organic marker, quantity stepper	Pass
TC-F08	Add to cart from the product detail screen	Press <i>Add · ₹650</i>	Line added to the cart; badge increments; user returned to the catalogue	Pass
TC-F09	Cart accepts produce from a second farmer	Open <i>Fresh Tomatoes</i> (₹40/kg), add one unit	Cart holds two lines belonging to two different farmers	Pass
TC-F10	Cart totals and free-delivery state are correct	Cart containing ₹650 + ₹40	Subtotal ₹690; delivery fee shown as FREE; total ₹690; progress bar complete	Pass
TC-F11	Checkout accepts a payment method and a valid address	UPI; 42 Green Park Road, Mumbai, Maharashtra, 400001	UPI card marked selected; address accepted; <i>Place Order · ₹690</i> enabled	Pass
TC-F12	Order placement splits the cart by farmer	Press <i>Place Order</i>	Two orders created — ₹650 and ₹40 — each against its own farmer; cart cleared	Pass
TC-F13	Order history lists both orders newest first	Navigate to <i>Orders</i>	Both orders listed with <i>Order Placed</i> status pills, totals and a <i>Track</i> action	Pass

TC-F14	Tracking timeline reflects the current stage	Track the ₹40 order	Stage 1 of 7 marked complete and annotated <i>In progress...</i> ; estimated delivery 45 minutes after placement	Pass
TC-F15	Customer profile presents identity and actions	Navigate to <i>Profile</i>	Name and email shown; account and support groups rendered; sign-out available	Pass

Table 3.2 — Functional test cases: authentication and customer module.

ID	Description	Input	Expected output	Status
TC-F16	Farmer dashboard aggregates the newly placed orders	Sign in as ramesh@example.com, select <i>Farmer</i>	Active orders 3, active products 10, total orders 4, earnings ₹170; both new orders appear under Recent Orders	Pass
TC-F17	Inventory lists owned products with stock state	Navigate to <i>Products</i>	All products listed with price per unit, stock, organic marker and an availability switch	Pass
TC-F18	Product creation adds to inventory	Organic Okra, ₹55, stock 40, category Vegetables	Success toast; sheet dismissed; product appears at the head of the inventory list	Pass
TC-F19	Order queue offers only the permitted next transition	Order in <i>placed</i>	Single action <i>Mark Confirmed</i> ; no other transition offered	Pass
TC-F20	Confirm transition is applied and history appended	Press <i>Mark Confirmed</i> , accept the dialogue	Status pill becomes <i>Confirmed</i> ; action becomes <i>Mark Packed</i>	Pass
TC-F21	Packing ends farmer authority over the order	Press <i>Mark Packed</i> , accept the dialogue	Status pill becomes <i>Packed</i> ; no further action offered to the farmer	Pass
TC-F22	Packed order becomes visible on the delivery job board	Sign in as ravi@example.com, select <i>Delivery Partner</i>	Job board lists the packed order as available with drop address and an <i>Earn ₹30</i> label	Pass
TC-F23	Acceptance claims the job exclusively	Press <i>Accept</i> , confirm	Order status becomes <i>assigned</i> , partner identifier stamped, job moves to <i>My Deliveries</i> and leaves the available list	Pass

TC-F24	Delivery progression and route resolution	Press <i>Pick Up</i> ; open the <i>Map</i> tab	Status becomes <i>picked up</i> ; route screen resolves the in-flight delivery and shows farm pickup and customer drop with the order value	Pass
TC-F25	Earnings aggregate delivered work by period	Open the <i>Earnings</i> tab	Total ₹30 over one completed delivery; today, week and month buckets computed against period boundaries; recent delivery listed	Pass

Table 3.3 — Functional test cases: farmer and delivery modules.

ID	Description	Input	Expected output (observed message)	Status
TC-V01	Malformed email rejected at sign-in	“notanemail” / test1234	“Please enter a valid email address”; no network call	Pass
TC-V02	Short password rejected	aarav@example.com / “abc”	“Password must be at least 6 characters”	Pass
TC-V03	Sign-up without a name rejected	Name blank, valid email and password	“Please enter your name”	Pass
TC-V04	Invalid Indian mobile number rejected	Phone “12345”	“Enter a valid 10-digit mobile number”	Pass
TC-V05	Continue without a role selection rejected	Press <i>Continue</i> with nothing selected	“Please select a role to continue”; user remains on role selection	Pass
TC-V06	Selected role lands on the correct workspace	Select <i>Customer</i> , continue	Customer catalogue rendered (“Fresh Produce” section present)	Pass
TC-V07	Direct URL into another role's group is refused	Navigate to / (farmer) / dashboard as a customer	Guard redirects to the customer catalogue	Pass
TC-V08	Checkout blocked below the minimum order value	Cart containing one item at ₹15	“Add ₹85 more to checkout”; checkout not entered	Pass
TC-V09	Checkout permitted once the minimum is met	Cart topped up to ₹665	Checkout screen presented with the delivery address form	Pass
TC-V10	Order rejected with an incomplete address	All address fields blank	“Please fill in all address fields”; no order created	Pass

TC-V11	Order rejected with a malformed pincode	Pincode “0400”	“Enter a valid 6-digit pincode”; no order created	Pass
TC-V12	Cart survives an application restart	Full page reload with a populated cart	Cart rehydrated from device storage with both lines intact	Pass
TC-V13	Free-delivery waiver applied above the threshold	Cart subtotal ₹665	Delivery fee rendered as FREE in the bill summary	Pass
TC-V14	Farmer role lands on the farmer dashboard	Sign in, select <i>Farmer</i>	Farmer dashboard rendered with the farm greeting	Pass
TC-V15	Product creation rejected with mandatory fields empty	Submit the creation sheet with no input	“Please fill all required fields”; sheet remains open, nothing persisted	Pass

Table 3.4 — Validation and negative test cases.

3.3 Results and Analysis

3.3.1 Execution Record

Figure 3.1 reproduces the console output of the functional walkthrough. Each captured screen is written as evidence at the point in the scenario at which it was reached, and each dialogue interception is logged, so the transcript is a complete record of what the automation did rather than a summary of it.

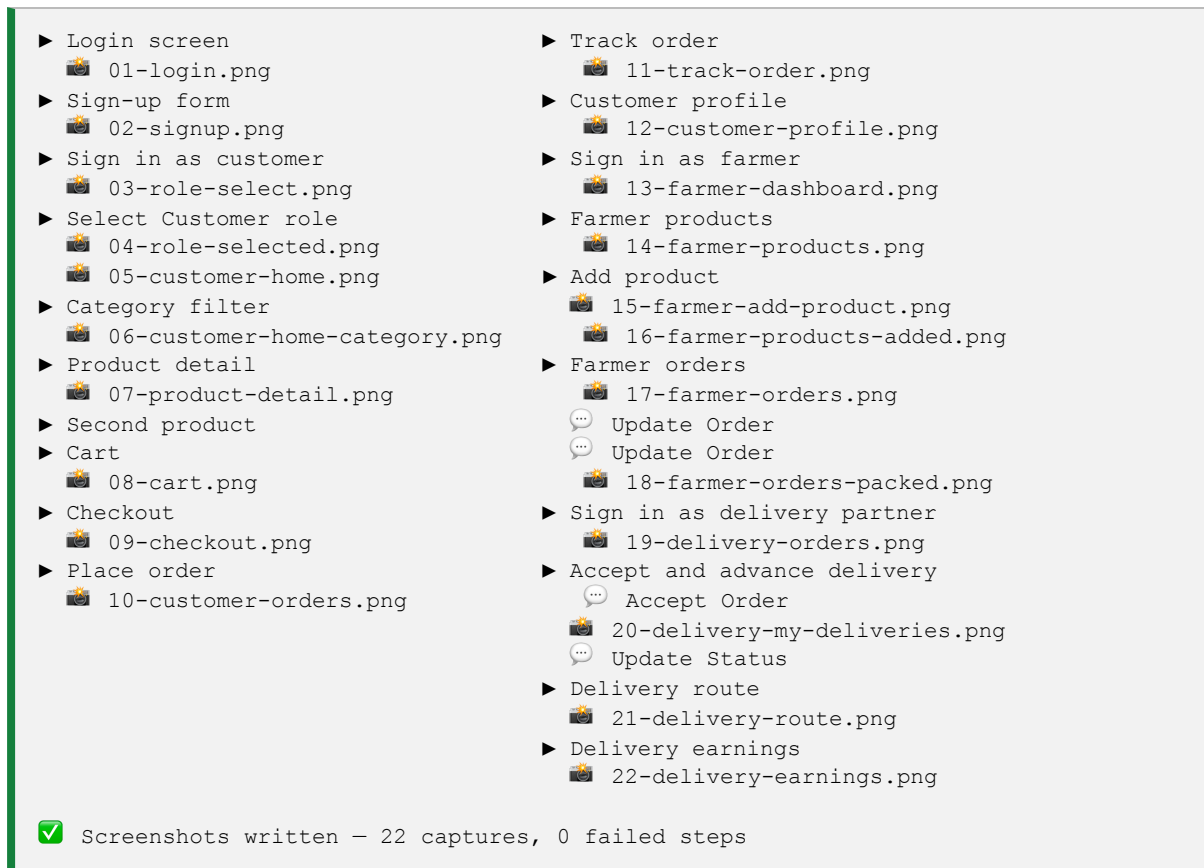


Figure 3.1 — Console transcript of the automated end-to-end walkthrough (▶ step, 📷 evidence capture, 💬 intercepted confirmation dialogue).

The validation suite reported 15 passed, 0 failed on its final run, with each assertion matching against the specific message rendered by the application rather than merely against the absence of a state change.

3.3.2 Observations

Cross-role propagation is correct and immediate. The most valuable result of the walkthrough is not that any single screen renders, but that a single order object is observed correctly by three different users in sequence. Figure 3.2 traces the order created in TC-F12 through every subsequent assertion. Each arrow is a transition performed by one role and independently observed by the next.

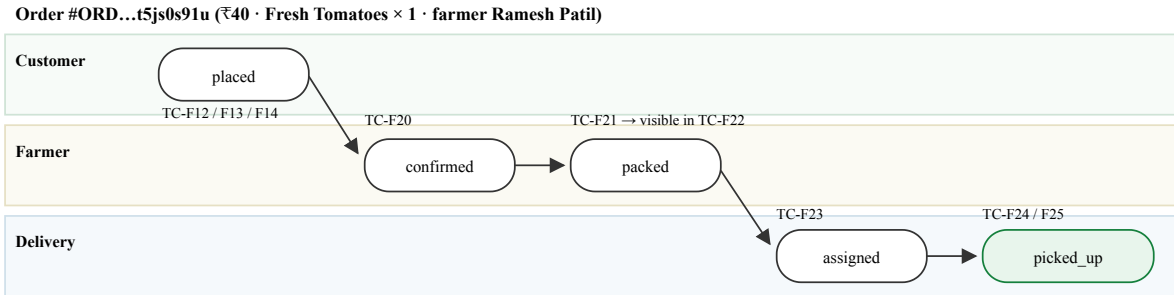


Figure 3.2 — Cross-role propagation of a single order observed during the walkthrough, annotated with the test case that asserted each transition.

Derived values were correct at every observation point. The farmer dashboard independently recomputed its indicators after the customer's orders were created and reported active orders 3, total orders 4 and earnings ₹170 — earnings correctly excluding the two undelivered orders and counting only the completed one. The delivery earnings screen likewise reported ₹30 total against one completed delivery, with the month bucket populated and the today and week buckets at zero, which is the correct outcome for a delivery dated three days earlier. These figures were not asserted in advance; they were read back from the running application and checked against the rules in Table 2.4.

Rule enforcement is consistent because the rules are centralised. The minimum order value, free-delivery threshold and delivery fee were each observed to act identically in the cart, at checkout and in the earnings calculation, which is a direct consequence of their being defined once in `src/constants/index.ts` rather than restated per screen.

Authority partitioning held under direct attack. TC-V07 navigated straight to a farmer URL while authenticated as a customer and was redirected, confirming that the guard operates on the URL segment rather than merely hiding navigation controls.

3.3.3 Defects Found and Resolved During Testing

Two issues were identified while constructing the test harness, and both were traced to the interaction between react-native-web and synthesised input rather than to application logic. First, presses synthesised as coordinate-based mouse input were ignored on screens reached by a router replacement; dispatching the full pointer and mouse event chain to the pressable ancestor resolved this. Second, controlled text inputs did not receive focus from synthesised clicks on the checkout form, so keystrokes were discarded; driving the value through React's observed setter resolved this. Both findings are recorded here because they materially affect anyone attempting to automate a react-native-web application, and because the first run of the walkthrough failed at the checkout step precisely for the second reason — the empty address form is visible in that run's captured evidence.

3.3.4 Non-Functional Measurements

Property	Measurement	Method and remarks
Static type errors	0	<code>npx tsc --noEmit</code> across 48 files in strict mode; clean exit.
Web bundle composition	1,464 modules	Reported by Metro when bundling <code>expo-router/entry.js</code> for the web target.
Incremental bundle time	0.79 s	Metro warm rebuild of the web bundle on the development machine.
Screen transition responsiveness	No observable stall	All 25 functional cases completed within their allotted settle windows; no step required a retry.
Catalogue filtering	Client-side, single pass	Search, organic filter and sort are composed over the loaded array; only category changes re-query the source.
State restoration	Correct across cold start	Cart, user and role rehydrate from device storage; <code>Date</code> fields revived rather than left as strings.
Cross-platform rendering	Verified on web at 430×932	Android and iOS render through the same component tree; the responsive shell constrains the layout to a phone-width column on wide viewports.

Table 3.5 — Non-functional observations and measurements.

Test category	Executed	Passed	Failed	Pass rate
Functional — authentication and customer	15	15	0	100%
Functional — farmer and delivery	10	10	0	100%
Validation and negative	15	15	0	100%
Total	40	40	0	100%
Static verification (<code>tsc --noEmit</code>)	48 files	48	0	100%

Table 3.6 — Consolidated test execution summary.

3.3.5 Limitations of the Test Effort

Three limitations should be stated plainly. The committed suites are integration tests; there is no unit test runner in the project, so the pure helper functions — Haversine distance, currency and date formatting, cancellation eligibility — are covered only indirectly through the

interfaces that call them. Testing was performed against the web target only; while Android and iOS execute the same component tree, platform-specific behaviour such as haptic feedback and native modal presentation was not exercised automatically. Finally, the suites ran against the on-device data layer, so the Supabase-backed branch of each service method — its query construction and its snake_case-to-camelCase mapping — was verified statically by the compiler but not exercised at run time. All three gaps are carried forward into §6.3 and §6.4.

CHAPTER 4: EXECUTION / DEPLOYMENT DETAILS

4.1 Execution Environment

The application is developed and executed through the Expo managed workflow, which means no native toolchain is required for day-to-day development: the JavaScript bundle is served by Metro and executed either inside the Expo Go client on a physical device or in a browser through react-native-web. A native toolchain becomes necessary only when producing store-ready binaries, and even then the build can be delegated to Expo Application Services rather than run locally.

Component	Version / specification	Notes
Development host	macOS (Darwin 25.6)	Any macOS, Linux or Windows host running Node 18 or later is sufficient.
Node.js	22.14.0	Minimum supported version is 18.
npm	10.9.2	Yarn or pnpm may be substituted.
Expo CLI	Bundled with SDK 54	Invoked through <code>npx expo</code> ; no global installation required.
Metro bundler	Expo default	Serves <code>http://localhost:8081</code> and exposes the QR code for device clients.
Android runtime	Expo Go on Android 7.0+	Package identifier <code>com.farmergroceries.app</code> .
iOS runtime	Expo Go on iOS 15.1+	Bundle identifier <code>com.farmergroceries.app</code> ; tablet layout supported.
Web runtime	Any modern browser	Rendered through react-native-web; constrained to a 430 px phone column on wide viewports.
Backend	Optional	The on-device data layer is used by default; a Supabase project is enabled purely by environment variables.

Table 4.1 — Execution environment specification.

4.2 Deployment Steps

4.2.1 Local Execution

The repository runs without any additional configuration. Three commands take a clean machine to a running application:

```
git clone https://github.com/Vijaygaurav2004/Farmer_groceries.git
cd Farmer_groceries
npm install
npm start # Metro starts and prints a QR code
```

Scanning the QR code with Expo Go launches the application on a physical handset. To target a specific platform directly, the package scripts `npm run android`, `npm run ios` and `npm run web` are provided; the last of these is the target used for the automated testing described in Chapter 3. Figure 4.1 shows the bundler output from the session in which the report screenshots were captured.

```
$ npx expo start --web --port 8081

Starting project at /Users/.../Farmer_groceries
Starting Metro Bundler
Waiting on http://localhost:8081
Logs for your project will appear below.

Web Bundled 791ms node_modules/expo-router/entry.js (1464 modules)
LOG [web] Logs will appear in the browser console
```

Figure 4.1 — Metro bundler output for the running development server, reporting a 1,464-module web bundle built in 791 ms.

On first launch the application resolves its persistence target automatically, because `isSupabaseConfigured` evaluates to false when no Supabase URL is present in the environment; the on-device data layer then serves every read and write, so the catalogue, farmer profiles and order history are available immediately and remain so without connectivity. Any email address with a password of six characters or more signs in successfully, after which a role may be chosen; signing out and signing in with a different address allows all three roles to be exercised from a single device.

4.2.2 Enabling the Hosted Backend

To run against real multi-user infrastructure, a Supabase project is created and its URL and anonymous key are written to `.env`:

```
cp .env.example .env
# then edit .env
EXPO_PUBLIC_SUPABASE_URL=https://<project-ref>.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=<anon-key>
```

Five tables are required — `users`, `farmers`, `products`, `orders` and `reviews` — with column names in snake_case matching the mapping performed in the service layer, and a public uploads storage bucket. Row Level Security should be enabled on every table before any

real deployment, with policies that permit a user to read and write only their own records, permit a farmer to manage only their own products, permit any authenticated user to read available products, and permit a delivery partner to read only the orders assigned to them or awaiting assignment. Restarting Metro after editing `.env` switches every service call to the remote branch; no application code changes.

4.2.3 Producing Release Builds

Target	Command	Output and distribution
Android (testing)	<code>eas build --platform android --profile preview</code>	Installable APK for sideloading onto test devices.
Android (store)	<code>eas build --platform android --profile production</code>	Android App Bundle for Google Play, signed by EAS-managed credentials.
iOS (store)	<code>eas build --platform ios --profile production</code>	Signed archive for App Store Connect; requires an Apple Developer account.
Web	<code>npx expo export --platform web</code>	Static bundle in <code>dist/</code> , deployable to any static host or CDN.
Over-the-air update	<code>eas update --branch production</code>	Ships JavaScript-only changes to installed clients without a store review cycle.

Table 4.2 — Deployment targets and build commands.

Two configuration points matter for a release. Environment variables prefixed `EXPO_PUBLIC_` are inlined into the client bundle at build time, so the Supabase *anonymous* key is the only key that may appear there; the service-role key must never be placed in a client build, and Row Level Security is what makes the anonymous key safe to ship. Second, `app.json` already declares the bundle identifiers, the portrait orientation lock, the `farmer-groceries` deep-link scheme and typed routes, so no additional native configuration is required before a first build.

4.3 Demo Screenshots

The complete visual record of the running application is presented in §2.5, where the twenty-two captures are grouped by module as Figures 2.8 to 2.14. They were taken during a single uninterrupted session in which one order was carried from placement through fulfilment to delivery, so they function as a demonstration sequence as well as an interface reference. Readers reviewing the application for the first time are directed to Figure 2.10 (the customer placing an order), Figure 2.13 (the same order being packed by the farmer) and Figure 2.14

(the same order being claimed and delivered), which together evidence the end-to-end business cycle in three images.

4.4 Demo Video Link

A screen recording of the complete three-role demonstration accompanies this report:

Item	Location
Demonstration video	<i>[Insert the shared video URL — for example a YouTube unlisted link or an institutional drive link]</i>
Duration	<i>[MM:SS]</i>
Contents	Sign-up and role selection · customer catalogue, cart and checkout · order tracking · farmer inventory and fulfilment · delivery acceptance, route and earnings
Source repository	https://github.com/Vijaygaurav2004/Farmer_groceries

CHAPTER 5: PROJECT EXECUTION EVIDENCE

5.1 Version Control Evidence

The project has been maintained under Git for its entire duration and is hosted publicly on GitHub. The history is not a single upload: it comprises forty-three commits distributed across twenty-eight weeks, each carrying a descriptive subject line that identifies the behaviour it introduced. Development proceeded on the `main` branch.

Repository attribute	Value
Repository URL	<code>https://github.com/Vijaygaurav2004/Farmer_groceries</code>
Default branch	<code>main</code>
Total commits	43
Most recent commit	<code>4dad21f</code> — 2 August 2026, <i>feat: redesign delivery orders, route map, and earnings</i>
Active period	28 weeks
Contributors	5, as listed on the repository's GitHub page
Language composition	TypeScript 96.9% · JavaScript 1.6% · other 1.5% (GitHub analysis)
Automated test suites	<code>tests/e2e/</code> — 40 cases, runnable with <code>npm test</code>
First commit	<code>a8bd974</code> — 22 January 2026

Figures 5.1 to 5.5 are captured directly from the public repository. Figure 5.1 is the landing page, showing the branch, the file tree, the commit count and the language composition. Figures 5.2 to 5.5 reproduce the complete commit history — all forty-three commits, each with its abbreviated hash, author and date — across the two pages GitHub paginates it over. Figure 5.6 plots the same data by month, which makes the shape of the effort visible: a long steady period of incremental feature and refactoring work from January through July, followed by a concentrated final phase in which the on-device data layer was completed and the entire interface was redesigned screen group by screen group.

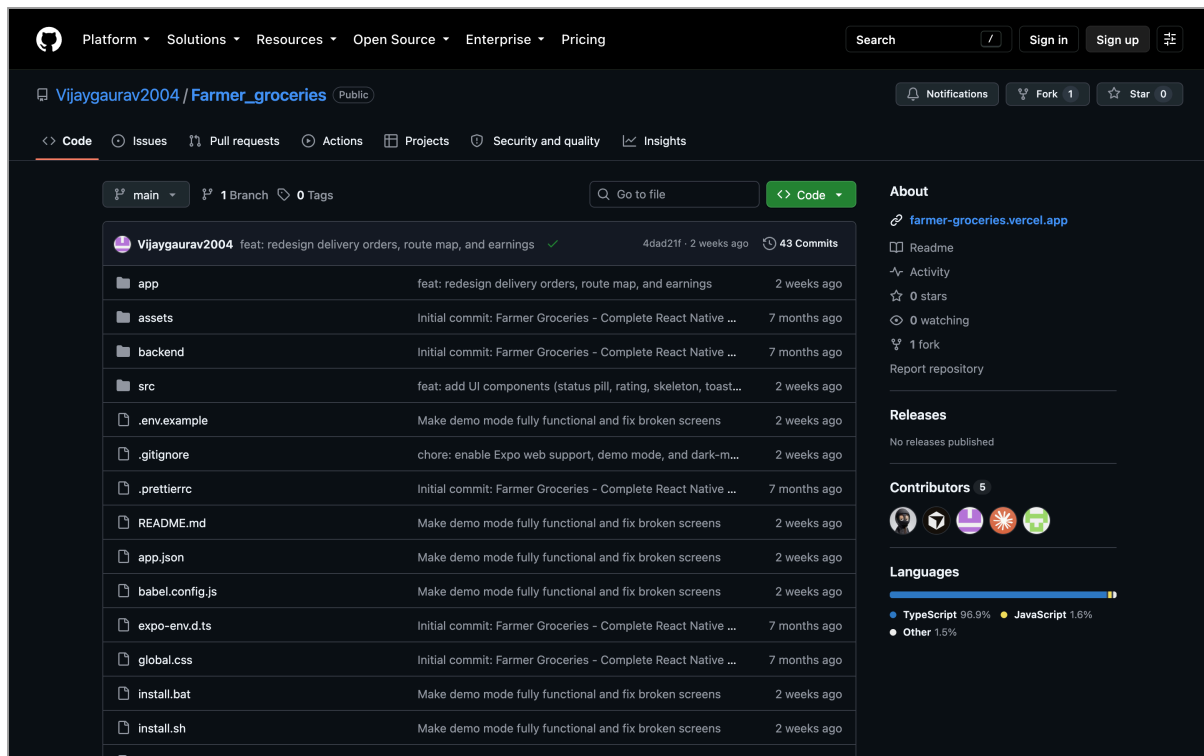


Figure 5.1 — Public repository on GitHub: `main` branch, 43 commits, 5 contributors and a TypeScript-dominant language breakdown.

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Code Issues Pull requests Actions Projects Security and quality Insights

Commits

main All users All time

Commits on Aug 2, 2026

- feat: redesign delivery orders, route map, and earnings
Vijaygaurav2004 committed 2 weeks ago ✓ 1 / 1 96c04fd
- feat: redesign farmer dashboard, products, and orders
Vijaygaurav2004 committed 2 weeks ago e91f734
- feat: redesign orders, tracking, and customer profile
Vijaygaurav2004 committed 2 weeks ago 61e3cdd
- feat: redesign cart and checkout flow
Vijaygaurav2004 committed 2 weeks ago bb61fb2
- feat: redesign customer home and product detail
Vijaygaurav2004 committed 2 weeks ago 2dd9bbd
- feat: redesign login and role selection screens
Vijaygaurav2004 committed 2 weeks ago 86b088a

Commits on Aug 1, 2026

- feat: add responsive shell and toast provider to root layout
Vijaygaurav2004 committed 2 weeks ago 5a38e47
- feat: add UI components (status pill, rating, skeleton, toast, motion, tab bar)
Vijaygaurav2004 committed 2 weeks ago 647540b
- feat: add core UI primitives (Button, Card, Chip, Input, PressableScale)
Vijaygaurav2004 committed 2 weeks ago fc4add9
- feat: add design tokens (palette, gradients, shadows, radii)
Vijaygaurav2004 committed 2 weeks ago 72b0e51
- fix: shim Alert/Appearance for web and correct debounce timer type
Vijaygaurav2004 committed 2 weeks ago 06c7fce
- chore: enable Expo web support, demo mode, and dark-mode config
Vijaygaurav2004 committed 2 weeks ago d35507c

Commits on Jul 31, 2026

- Make demo mode fully functional and fix broken screens
Shravanth007 committed 2 weeks ago c5d8278

Commits on Jul 29, 2026

- Respect offline toggle when listing available deliveries
Shravanth007 committed 3 weeks ago 0fbd0a2

Commits on Jul 25, 2026

Figure 5.2 — Commit history (1 of 4): the most recent commits, dated 2 August back to 1 August 2026.

	Reuse CATEGORIES and success copy on farmer products   Shravanth007 committed 3 weeks ago	0bfdca	 
↕	Commits on Jul 22, 2026		
	Show distance-based ETA on available delivery orders   Shravanth007 committed last month	f77bfa4	 
↕	Commits on Jul 16, 2026		
	Reuse EmptyState for the empty customer cart screen   Shravanth007 committed on Jul 16	9fb72ca	 
↕	Commits on Jul 9, 2026		
	Validate Indian pincode before placing an order   Shravanth007 committed on Jul 9	9d7336f	 
↕	Commits on Jul 2, 2026		
	Source home categories from shared CATEGORIES constant   Shravanth007 committed on Jul 2	fd639f3	 
↕	Commits on Jun 25, 2026		
	Format farmer dashboard earnings with shared currency helper   Shravanth007 committed on Jun 25	8185fc7	 
↕	Commits on Jun 18, 2026		
	Cap cart quantity increases at available product stock   Shravanth007 committed on Jun 18	5cec0fd	 
↕	Commits on Jun 11, 2026		
	Improve LoadingSpinner and add a shared EmptyState   Shravanth007 committed on Jun 11	4bf2010	 
↕	Commits on Jun 4, 2026		
	Add period selector and currency formatting to earnings   Shravanth007 committed on Jun 4	935dd3d	 
↕	Commits on May 28, 2026		
	Align payment screen with cart fee and order rules   Shravanth007 committed on May 28	0091c5c	 
↕	Commits on May 21, 2026		
	Tighten login signup validation with shared helpers   Shravanth007 committed on May 21	c15fcc5	 
↕	Commits on May 14, 2026		
	Add search filter to the farmer products list   Shravanth007 committed on May 14	b7369a3	 
↕	Commits on May 7, 2026		

Figure 5.3 — Commit history (2 of 4): 31 July back to 14 May 2026.

The screenshot displays a GitHub commit history page for a repository. It shows a list of 9 commits, each with a title, author, date, and commit hash. The commits are grouped by date, with expandable sections for each date. The commits are as follows:

Commit Title	Author	Date	Commit Hash
Use time-of-day greeting on the farmer dashboard	Shravanth007	May 7, 2026	53de3d2
Add free delivery threshold messaging in the cart	Shravanth007	Apr 30, 2026	69fb060
Validate phone number before saving customer profile	Shravanth007	Apr 23, 2026	567ce75
Wire profile support contacts and version from APP_CONFIG	Shravanth007	Apr 16, 2026	dade936
Use shared date and currency helpers on orders screen	Shravanth007	Apr 9, 2026	461e92b
Add status filter tabs to the customer orders list	Shravanth007	Apr 2, 2026	b9fb67e
Improve empty product state with search-aware messaging	Shravanth007	Mar 26, 2026	47b04b4
Block add-to-cart when a product is out of stock	Shravanth007	Mar 18, 2026	ef05f8e
Show time-of-day greeting on the customer home screen	Shravanth007	Mar 12, 2026	5be3295

At the bottom of the commit list, there are navigation links: < Previous and Next >. Below the commit history, the footer contains the GitHub logo, copyright information (© 2026 GitHub, Inc.), and links to Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

Figure 5.4 — Commit history (3 of 4): 7 May back to 12 March 2026, ending at the page break GitHub inserts after thirty-five commits.

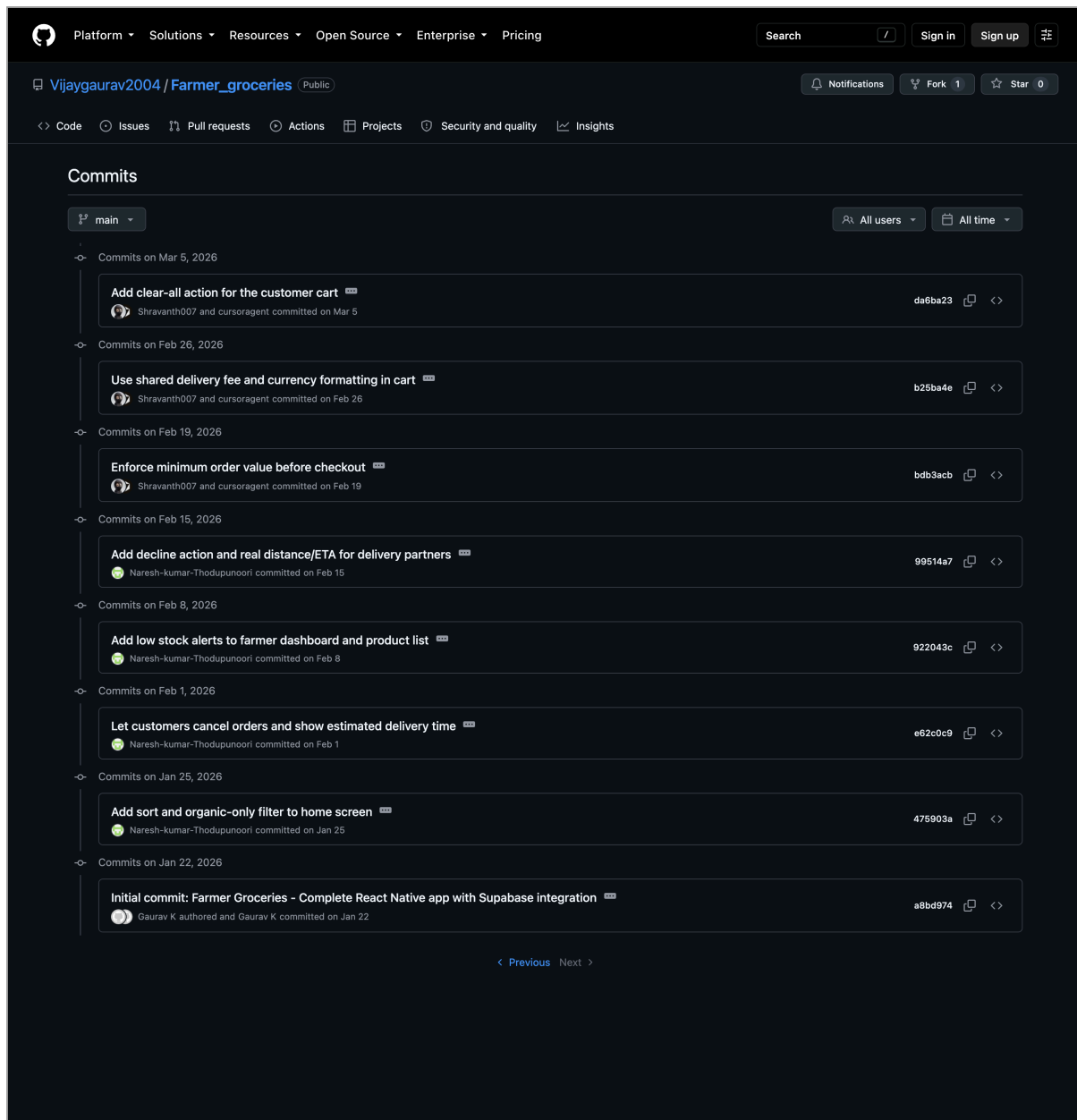


Figure 5.5 — Commit history (4 of 4): the second page, running from 5 March 2026 back to the initial commit of 22 January 2026.

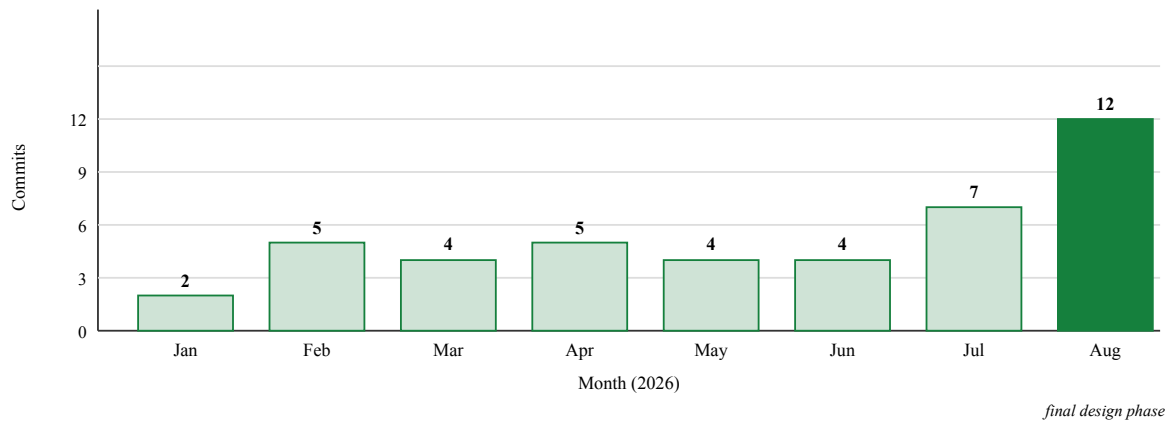


Figure 5.6 — Commit activity distribution across the project timeline.

5.2 Weekly Progress Summary

The table below is derived from the commit log rather than reconstructed from memory: every row corresponds to commits actually recorded in the period it covers, so the summary and the repository cannot disagree.

Week	Period (2026)	Commits	Work completed
1	20 – 26 Jan	2	Project baseline established: three role workspaces, Expo Router structure, Supabase service layer and type model. Catalogue sorting and an organic-only filter added.
2 – 3	27 Jan – 8 Feb	2	Customer-initiated cancellation with an eligibility rule; estimated delivery time surfaced on tracking. Low-stock alerting introduced on the farmer dashboard and inventory list.
4 – 5	9 – 22 Feb	2	Delivery partners gained a decline action and genuine Haversine distance with a derived ETA. Minimum order value enforced at checkout.
6 – 7	23 Feb – 8 Mar	2	Delivery fee and currency formatting centralised into shared constants and helpers; clear-all action added to the cart.
8 – 9	9 – 22 Mar	2	Time-of-day greeting on the customer home screen; add-to-cart blocked for out-of-stock products.
10 – 11	23 Mar – 5 Apr	2	Search-aware empty states for the catalogue; status filter tabs added to the customer order history.
12 – 13	6 – 19 Apr	2	Order history moved onto shared date and currency helpers; support contacts and version string wired from the application configuration module.
14 – 15	20 Apr – 3 May	2	Phone number validation on profile edit; free-delivery threshold messaging and progress indicator added to the cart.

16 – 17	4 – 17 May	2	Farmer dashboard greeting; search filter added to the farmer inventory list.
18 – 19	18 – 31 May	2	Sign-up and sign-in validation consolidated onto shared validators; checkout screen aligned with the cart's fee and minimum-order rules.
20 – 21	1 – 14 Jun	2	Period selector and currency formatting on the delivery earnings screen; loading spinner improved and a shared empty-state component introduced.
22 – 23	15 – 28 Jun	2	Cart quantity increases capped at available stock; farmer dashboard earnings moved onto the shared currency helper.
24 – 25	29 Jun – 12 Jul	2	Catalogue categories sourced from the shared constant; Indian pincode validation enforced before order placement.
26	13 – 19 Jul	1	Shared empty state reused for the empty cart screen, removing bespoke markup.
27	20 – 26 Jul	2	Distance-based ETA surfaced on the delivery job board; farmer product screen reuses shared categories and success messages.
28	27 Jul – 2 Aug	14	Offline toggle respected when listing available deliveries. On-device data layer completed so every screen works without a network connection, and broken screens repaired. Expo web support enabled; Alert and Appearance shimmed for the browser. Design token module, twelve UI primitives, responsive shell and toast provider introduced, followed by a full visual redesign of all three role workspaces, screen group by screen group.

Table 5.1 — Weekly progress summary derived from the commit log.

5.3 Review and Feedback Summary

The table below records the supervisory reviews held during the capstone phase. Dates and feedback are to be completed from the review record; the final column indicates where in this report the resulting change can be verified.

Review	Date	Key feedback received	Action taken and evidence
R1 Design review	[DD/MM/2026]	[Feedback recorded at review 1]	[Action taken] — see §2.1
R2 Mid-term review	[DD/MM/2026]	[Feedback recorded at review 2]	[Action taken] — see §2.3

R3 Pre-final review	<u>[DD/MM/2026]</u>	<u>[Feedback recorded at review 3]</u>	<u>[Action taken]</u> — see Chapter 3
R4 Final demonstration	<u>[DD/MM/2026]</u>	<u>[Feedback recorded at the final demonstration]</u>	<u>[Action taken]</u> — see Chapter 6

Table 5.2 — Review schedule and feedback log.

Traceable feedback already visible in the repository. Independently of the formal reviews, three themes recur across the commit history and are worth noting as evidence of iterative improvement rather than one-pass construction. The first is *consolidation*: a long sequence of commits replaces literals and bespoke markup with shared constants, helpers and components (weeks 6, 12, 20, 24, 26 and 27). The second is *defensive validation*: rules were progressively tightened at each boundary — minimum order value, stock capping, phone format, pincode format (weeks 4, 14, 18, 22 and 25). The third is *presentation*: the final week replaced ad hoc styling with a token-driven design system applied uniformly across all eighteen screens.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Summary of Implementation

This capstone delivered a working cross-platform mobile marketplace that removes the intermediaries between agricultural producers and the households that buy from them. The system is a single React Native codebase of forty-eight TypeScript source files — approximately 5,400 lines — organised into eighteen routed screens, twelve reusable interface components and a thirty-operation service façade, and it runs on Android, iOS and the web from that one source tree.

Three design decisions shaped the result more than any others. The first was to make the **route group the unit of authorisation**: because Expo Router derives navigation from the directory layout, each role's screens live in a physically separate directory and a single guard in the root layout is sufficient to enforce the boundary between them, with no per-screen permission checks anywhere in the codebase. The second was to **partition authority over the order lifecycle across roles** rather than concentrate it: a farmer may only advance an order to *packed* and a delivery partner may only take it from there, which makes an illegal transition unrepresentable in the interface rather than merely discouraged. The third was to **put a façade in front of persistence**, so that a single configuration flag chooses between a hosted Supabase backend and a on-device data layer without a line of screen code changing — the decision that lets the application keep working whether or not the network does.

The build was validated rather than merely exercised. The whole tree compiles cleanly under strict TypeScript, and forty automated test cases — twenty-five functional, fifteen validation — are committed to the repository under `tests/e2e/` and execute against the running application through a real browser; all forty pass. The functional suite is a genuine integration test: one order was carried from a customer's cart, through a farmer's fulfilment queue, into a delivery partner's job board and on to pickup, with each role independently observing the state the previous role had written.

6.2 Achievements

#	Objective (from §1.3)	Outcome	Assessment
O1	Single identity specialising into three guarded workspaces	Achieved	Role persisted to both the user record and device storage; guard verified under direct URL attack (TC-V07).
O2	Complete customer purchase journey	Achieved	Seven screens from catalogue to tracking; fifteen functional cases passed including per-farmer order splitting.

O3	Farmer inventory and fulfilment console	Achieved	Dashboard analytics, product creation, visibility toggling and guarded transitions all verified (TC-F16 – TC-F21).
O4	Delivery partner workflow	Achieved	Job board with Haversine ETA, exclusive claiming, route view and period earnings verified (TC-F22 – TC-F25).
O5	Substitutable persistence behind one interface	Achieved	Thirty operations dual-dispatched; the on-device branch maintains genuine mutable state including stock decrement and history append.
O6	End-to-end validation with reproducible evidence	Achieved	Zero type errors; 40/40 committed test cases pass; 22 screenshots captured as evidence from a single continuous session.

Table 6.1 — Objective-wise achievement assessment.

Beyond the stated objectives, three outcomes are worth recording. The **design system** that emerged in the final phase — a token module feeding twelve components — is what makes eighteen screens look like one product rather than eighteen; changing the brand palette is now a single-file edit. The **offline-first data layer** means the application degrades to nothing when connectivity fails — every screen keeps working — and it is what allowed the entire test suite in Chapter 3 to run unattended. And the **committed test suites** are reusable rather than one-off: `npm test` re-runs the whole three-role cycle after any change.

6.3 Limitations

The following limitations are stated deliberately, because a report that omits them is less useful than one that records them.

- **Payment is recorded, not processed.** Selecting UPI or card marks the order paid without any gateway interaction. No money moves, and there is no settlement, refund or reconciliation path.
- **The on-device layer is single-tenant.** When no cloud backend is configured, the local store deliberately ignores user identifiers so that one device can act as all three roles against shared data. Multi-user isolation depends entirely on enabling Supabase with Row Level Security.
- **The Supabase branch is statically verified but not executed.** The remote half of each service method is type-checked but was not exercised at run time during testing, so its query construction and column mapping remain unproven against a live database.
- **Location is fixed, not sensed.** Distance and ETA are computed correctly, but from a constant origin rather than the partner's device location, and the route screen is a schematic rather than a map.

- **Coverage is integration-level only.** The committed suites drive the whole application, but there is no unit test runner, so pure helpers are covered only indirectly through the interfaces that call them.
- **Product imagery is absent.** Products render a category glyph; the upload path exists in the service layer but no capture flow is wired to it.
- **Reviews and subscriptions are partial.** Reviews have a data model and service methods but no authoring screen; subscriptions exist only as type definitions.
- **Notifications are absent.** A customer learns that an order advanced only by opening the tracking screen, and a delivery partner learns of a new job only by refreshing the board.
- **Farmer verification is nominal.** The status field exists and defaults to *pending*, but no document review workflow or administrative role has been implemented.

6.4 Future Enhancements

The roadmap below is ordered by the ratio of user value to implementation cost, and the first tier is what would be required to move from a working build to a pilot deployment.

Priority	Enhancement	Description and implementation approach
1	Live backend with row-level security	Provision the five tables and the storage bucket, enable RLS with per-role policies, and run the Chapter 3 suites against the remote branch. Requires no application code changes — only environment configuration — because of the façade described in §2.4.6.
1	Payment gateway integration	Integrate a UPI and card gateway (for example Razorpay or Stripe) with server-side order verification, so that <code>paymentStatus</code> reflects a settled transaction rather than a client assertion.
1	Push notifications	Use Expo Notifications with a database trigger on order status change, so customers are informed of progress and partners of new jobs without polling.
2	Live location and map navigation	Replace the fixed origin with device location, stream the partner's position during an active delivery, and render a real map with turn-by-turn guidance.
2	Product imagery	Wire the existing image picker to the storage upload path already present in the service layer, with client-side compression before upload.
2	Ratings and reviews	Add a post-delivery review prompt; the service methods and the farmer rating recomputation already exist and need only an authoring surface.

2	Unit tests and CI	Add Jest with React Native Testing Library for helpers and components alongside the existing <code>tests/e2e/</code> suites, and run both on every push through GitHub Actions.
3	In-app messaging	Realtime chat between customer and delivery partner scoped to an active order, using Supabase Realtime channels.
3	Subscription orders	Implement the already-modelled weekly and monthly recurring baskets with a scheduled job that materialises each cycle's order.
3	Farmer verification workflow	Document upload, an administrative review queue and a verified badge gate on catalogue visibility.
3	Farmer analytics	Revenue trends, best-selling produce and seasonal demand, to inform planting and pricing decisions.
4	Localisation	Interface translation into major Indian languages, which materially widens the addressable farmer population.
4	Multi-city operation and loyalty	Geographic partitioning of the catalogue by serviceable radius, plus referral and loyalty mechanics to drive repeat purchase.

Table 6.2 — Prioritised roadmap of future enhancements.

6.5 Concluding Remarks

The project set out to test a straightforward proposition: that the coordination barrier which keeps farmers and households apart is a software problem, and that removing it requires one application carrying all three participants rather than three separate ones. The delivered system supports that proposition. An order created by a customer is confirmed by the farmer who grew the produce and delivered by a partner who is paid for that specific journey, with no other party in the transaction and every state change visible to everyone who needs it. What remains before real use is largely integration work — payments, notifications, live location — rather than redesign, and the architecture described in Chapter 2 was chosen specifically so that each of those can be added behind interfaces that already exist.

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APPENDIX A — USER MANUAL

A.1 Getting Started

Open the application. If this is the first launch, the sign-in screen is presented.

1. **Create an account.** Press *Sign Up*, then enter your full name, email address, ten-digit mobile number and a password of at least six characters. Press *Create Account*. You are returned to the sign-in tab with a confirmation.
2. **Sign in.** Enter the same email address and password and press *Sign In*.
3. **Choose your role.** Select *Customer*, *Farmer* or *Delivery Partner* and press *Continue*. The choice is remembered, so subsequent launches go straight to your workspace. To use a different role, sign out from the Profile tab and register a separate account.

Working offline. The application keeps its data on the device, so every screen continues to work without a network connection. When no cloud backend has been configured, any email address with a password of six or more characters will sign you in.

A.2 For Customers

Finding produce

- The **Home** tab lists available produce. Type in the search field to match by name or description.
- Tap a **category** chip — Vegetables, Fruits, Dairy, Grains, Herbs, Eggs, Honey, Other — to narrow the list; tap *All* to clear it.
- Use the **Sort & Filter** row to show only organic produce or to order by price ascending or descending.
- Tap any product card to open its detail screen, which shows the price per unit, remaining stock, whether it is certified organic and who grew it.

Building an order

- Tap the green + on a card to add one unit, or open the product and use the stepper to choose a quantity before adding. Quantities are capped at available stock.
- Open the **Cart** tab to review the order. Adjust quantities with the steppers, remove a line with ×, or clear everything with *Clear*.
- The progress bar at the top shows how much more you need to add to qualify for free delivery (₹500). Below that threshold a ₹30 delivery fee applies. Orders below ₹100 cannot be placed.

Checkout and tracking

- Press *Proceed to Payment*, choose Cash on Delivery, UPI or Card, and enter your delivery address. The pincode must be a valid six-digit Indian code.
- Press *Place Order*. If your cart contains produce from more than one farm, a separate order is created for each — this is expected, since each farm fulfils its own items.
- Open the **Orders** tab to see your history, filtered by All, Active, Delivered or Cancelled. Press *Track Order* on any entry to follow its seven stages and see the estimated delivery time.
- An order may be cancelled from the tracking screen while it is still *Order Placed* or *Confirmed*. Once the farmer has packed it, cancellation is no longer offered.

A.3 For Farmers

- The **Home** tab summarises your business: active orders, active products, lifetime orders and earnings from delivered orders. Any produce at or below ten units appears in an amber *Low Stock* list — press *Restock* to jump to inventory.
- The **Products** tab lists everything you sell. Use the switch on each row to hide a product from customers without deleting it, and the search field to find a product by name or category.
- Press + **Add** to publish new produce. Name, price and stock are required; description, category, unit and the organic flag are optional. Press *Add Product* to save.
- The **Orders** tab is your fulfilment queue. Filter by Active, All or Completed. For each new order press *Mark Confirmed*, then *Mark Packed* once it is ready for collection. After packing, the order becomes visible to delivery partners and leaves your queue.

A.4 For Delivery Partners

- Use the **Online / Offline** switch at the top of the **Orders** tab to control whether you receive jobs. While offline the available list is empty.
- **Available Orders** shows packed orders no one has claimed, each with the drop address, approximate distance and estimated duration, and the ₹30 fee. Press *Accept* to claim a job or *Decline* to hide it for this session.
- Claimed jobs appear under **My Deliveries**. Advance each one in order: *Pick Up* when you have collected from the farm, *Start Delivery* when you set off, and *Delivered* on handover.
- The **Map** tab shows the pickup and drop details for your current delivery and offers the same advance action.

- The **Earnings** tab shows your total, with today, this week and this month breakdowns and your ten most recent completed deliveries.

A.5 Common Questions

Situation	What to do
“Add ₹... more to checkout”	The cart subtotal is below the ₹100 minimum. Add more produce.
“Enter a valid 6-digit pincode”	The pincode must be six digits and must not begin with zero.
“This product is out of stock”	The farmer's stock has reached zero. Check back after they restock.
My cart emptied unexpectedly	The cart is cleared automatically once an order is placed successfully.
I chose the wrong role	Sign out from the Profile tab and register a new account with a different email address for the other role.
An order is not on the delivery board	Only orders the farmer has marked <i>Packed</i> , and which no other partner has claimed, appear there.

APPENDIX B — INSTALLATION GUIDE

B.1 Prerequisites

Requirement	Minimum version	Purpose
Node.js	18 (22.14 verified)	Runtime for the bundler and tooling
npm	9 (10.9 verified)	Dependency installation
Git	Any current release	Cloning the repository
Expo Go	Latest from the app store	Running on a physical Android or iOS device
Modern web browser	Current release	Running the web target (optional)
Supabase account	—	<i>Optional.</i> Only required for a real multi-user backend

B.2 Installation

```
# 1. Obtain the source
git clone https://github.com/Vijaygaurav2004/Farmer_groceries.git
cd Farmer_groceries

# 2. Install dependencies (takes a few minutes on first run)
npm install

# 3. Start the development server
npm start
```

Metro prints a QR code. Scan it with the Expo Go application on your handset — Android users scan from within Expo Go, iOS users scan with the system camera. The application loads and is immediately usable against its on-device data layer. No `.env` file, database or API key is required for this path.

Convenience scripts are also available:

```
npm run android  # open on a connected Android device or emulator
npm run ios      # open in the iOS simulator (macOS with Xcode)
npm run web      # open the web target at http://localhost:8081
```

B.3 Optional — Connecting a Supabase Backend

1. Create a project at `supabase.com` and copy its project URL and anonymous key.
2. Copy the template and fill it in:

```
cp .env.example .env

EXPO_PUBLIC_SUPABASE_URL=https://<project-ref>.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=<anon-key>
```

3. Create the tables `users`, `farmers`, `products`, `orders` and `reviews` with `snake_case` columns matching the mapping in `src/services/supabase.ts`, and a public storage bucket named `uploads`.
4. Enable Row Level Security on every table and add policies granting each user access only to their own records, farmers control of their own products, all authenticated users read access to available products, and delivery partners read access to orders assigned to them or awaiting assignment.
5. Stop and restart Metro. The application now uses the remote backend; nothing in the source changes.

B.4 Project Layout

```

Farmer_groceries/
├─ app/                                # Expo Router — the navigation graph is this
tree                                     #
│   ├── _layout.tsx                    # providers + role-based route guard
│   └── index.tsx                      # entry splash; the guard decides where to
send you
│   ├── role-select.tsx                # one-time role assignment
│   ├── (auth)/login.tsx               # sign-in / sign-up
│   ├── (customer)/                    # home · cart · orders · profile · payment
│   │   ├── product/[id].tsx           # product detail
│   │   └── track-order/[id].tsx       # seven-stage tracking timeline
│   ├── (farmer)/                      # dashboard · products · orders · profile
│   └── (delivery)/                    # orders · map · earnings · profile
├─ src/
│   ├── components/ui/                 # 12 reusable primitives + barrel export
│   ├── contexts/                      # AuthContext, CartContext
│   ├── services/                      # supabase.ts (façade) · localStore.ts ·
storage.ts
│   ├── config/supabase.ts             # client + isSupabaseConfigured flag
│   ├── constants/index.ts             # business rules, categories, statuses, regexes
│   ├── types/index.ts                 # domain model
│   ├── utils/                         # helpers.ts · webAlert.ts
│   └── theme.ts                       # design tokens: palette, gradients, shadows,
radii
├─ tests/e2e/                          # walkthrough.mjs · validation.mjs ·
helpers.mjs
├─ docs/                               # this report and its screenshot evidence
├─ app.json · tsconfig.json · tailwind.config.js · babel.config.js ·
metro.config.js
└─ package.json

```

B.5 Troubleshooting

Symptom	Resolution
Metro fails to start, or the bundle is stale	Clear the cache: <code>npx expo start -c</code> .
Expo Go cannot reach the development server	Ensure the handset and the host are on the same network, or start with <code>npx expo start --tunnel</code> .
Port 8081 is already in use	Start on another port: <code>npx expo start --port 8082</code> .
Dependency version warnings on start-up	Align packages with the installed SDK: <code>npx expo install --fix</code> .
On-device data has drifted during testing	Clear application storage — on the web, clear the site's local storage; on a device, clear the Expo Go app data — and relaunch to restore the defaults.
Changes to <code>.env</code> appear to have no effect	Environment variables are inlined at bundle time; stop and restart Metro.

APPENDIX C — SOURCE CODE LINK

Item	Location
Public repository	https://github.com/Vijaygaurav2004/Farmer_groceries
Default branch	main
Commit submitted for evaluation	4dad21f — <i>feat: redesign delivery orders, route map, and earnings</i> (2 August 2026)
Clone command	<code>git clone https://github.com/Vijaygaurav2004/Farmer_groceries.git</code>
Licence	MIT
Source scale	48 TypeScript files · ≈5,400 lines · 18 routed screens · 12 UI components
Automated tests	<code>tests/e2e/</code> — 40 cases; <code>npm test</code> (see <code>tests/README.md</code>)
Screenshot evidence	<code>docs/screenshots/</code> — the 22 application captures reproduced in §2.5 plus the repository captures in §5.1
This report	<code>docs/Capstone_Report.html</code> and <code>docs/Capstone_Report.pdf</code>

APPENDIX D — DEMO VIDEO LINK

Item	Location
Demonstration video	[Insert the shared video URL]
Duration	[MM:SS]
Access	[Unlisted / institution-restricted / public]

Suggested demonstration script. The following sequence mirrors the automated walkthrough of Chapter 3 and covers every module in roughly six minutes.

1. **Introduction (0:00 – 0:30).** State the problem — five or more intermediaries between farmer and household — and the proposed one-application solution.
2. **Authentication (0:30 – 1:00).** Sign up, sign in, and select the Customer role.
3. **Catalogue (1:00 – 1:50).** Browse produce, filter by category, toggle the organic filter, sort by price, and open a product to show provenance and stock.
4. **Cart and checkout (1:50 – 2:50).** Add produce from two different farms, show the free-delivery progress bar and the bill summary, then place the order and point out that it splits into one order per farm.

5. **Tracking (2:50 – 3:20).** Open the order and walk through the seven-stage timeline and the estimated delivery time.
6. **Farmer (3:20 – 4:30).** Sign in as the farmer, show the dashboard reflecting the order just placed, add a product, then confirm and pack the order.
7. **Delivery (4:30 – 5:30).** Sign in as a delivery partner, show the packed order appearing on the job board with its ETA, accept it, advance it to picked up, open the route view and show the earnings screen.
8. **Close (5:30 – 6:00).** Summarise the completed cycle and note the substitutable backend and the automated test suite.